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Decoding vision transformer variations for image classification: A guide to performance and usability

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ABSTRACT

With the rise of Transformers, Vision Transformers (ViTs) have become a new standard in visual recognition. This has led to the development of numerous architectures with diverse designs and applications. This survey identifies 22 key ViT and hybrid CNN–ViT models, along with 5 top Convolutional Neural Network (CNN) models. These were selected based on their new architecture, relevance to benchmarks, and overall impact. The models are organised using a defined taxonomy formed by CNN-based, pure Transformer-based, and hybrid architectures. We analyse their main components, training methods, and computational features, while assessing performance using reported results on standard benchmarks such as ImageNet and CIFAR, along with our training and fine-tuning evaluations on specific imaging datasets. In addition to accuracy, we look at real-world deployment issues by analysing the trade-offs between accuracy and efficiency in embedded, mobile, and clinical settings. The results indicate that modern CNNs are still very competitive in limited-resource environments, while advanced ViT variants perform well after large-scale pretraining, especially in areas with high variability. Hybrid CNN–ViT architectures, on the other hand, tend to offer the best balance between accuracy, data efficiency, and computational cost. This survey establishes a consolidated benchmark and reference framework for understanding the evolution, capabilities, and practical applicability of contemporary vision architectures.

1. Introduction

Vision Transformer (ViT) represents a significant shift in the landscape of computer vision, greatly supported by the transformer architecture, which was initially designed for natural language processing tasks. Introduced by Dosovitskiy et al. (2020), ViTs have become one of the most popular architectures to employ, outperforming traditional Convolutional Neural Networks (CNNs) models on many image recognition tasks, particularly when the model is pre-trained on large datasets.

ViTs leverage the self-attention mechanism to capture long-range dependencies and global context across the entire image. Capturing global feature context enables learning the relationship between image regions, contrary to the local receptive fields of CNNs, allowing for the creation of more effective models that can accommodate more complex datasets (Yuan et al., 2021), thereby achieving state-of-the-art performance.

The transformer architecture is highly modular and flexible (Touvron et al., 2021), making it easier to adapt and extend for various vision tasks beyond image classification, such as object detection and segmentation, by reusing the transformer encoder components. To address these issues, robust initialisation techniques combined with adaptive optimisation algorithms can simplify training and reduce the dependence on large-scale datasets.

Training ViTs presents unique challenges. Effective fine-tuning depends on access to high-quality pre-trained backbone models, since training from scratch typically requires careful tuning of numerous hyperparameters (Tian et al., 2025; Wang et al., 2022; Zhai et al., 2022) and large datasets to optimise learning (Steiner et al., 2021). ViTs are also highly sensitive to parameter choices, such as learning rates and regularisation strategies (Wang et al., 2022), which complicates deployment.

Different ViTs variations have been proposed to address these difficulties by optimising the architecture with the introduction of lightweight or efficient transformer models tailored for resource-constrained environments. To overcome some of the problems introduced by ViT, hybrid architectures that combine ViTs with CNNs have

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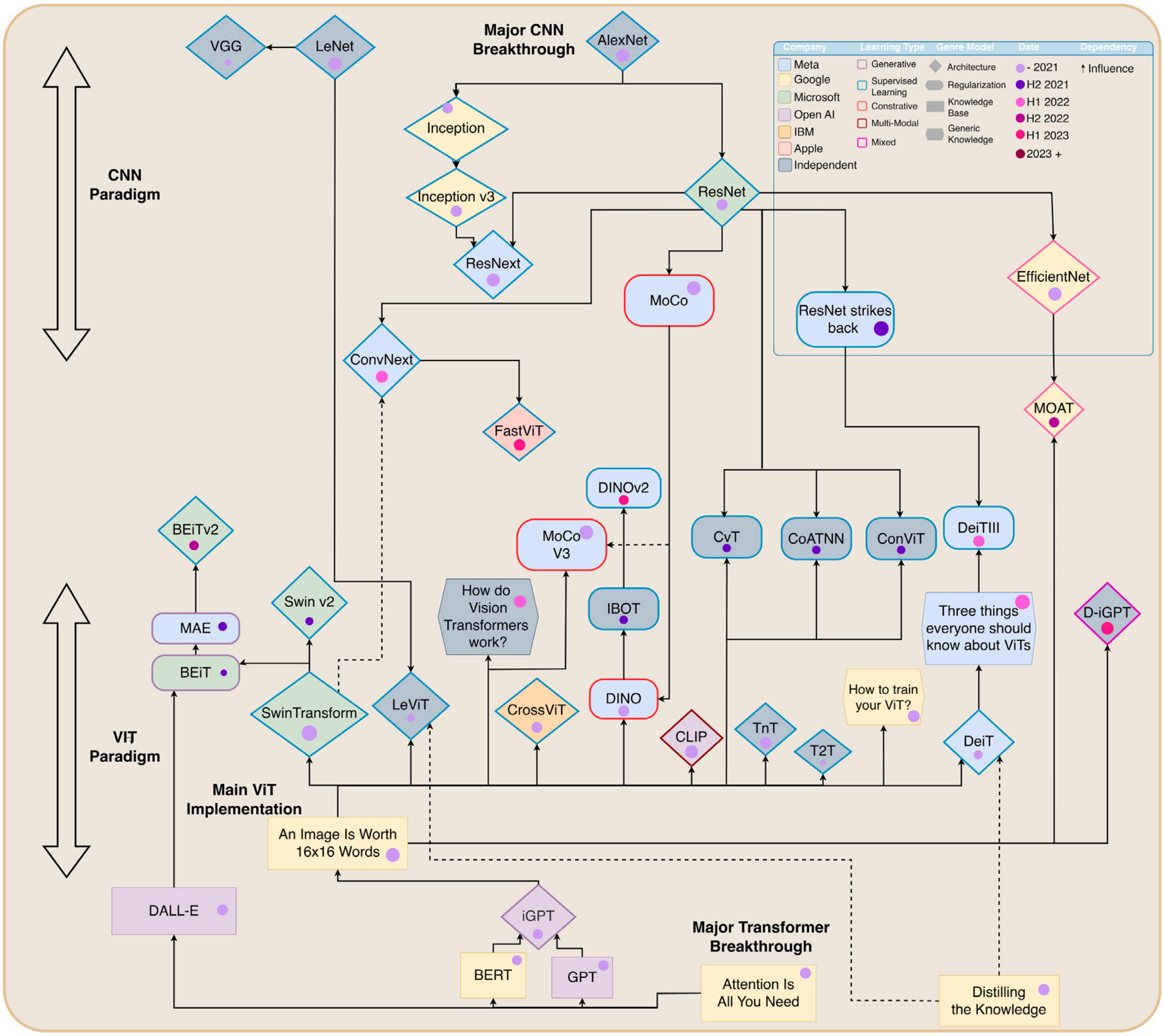

Fig. 1. Diagram illustrating the landscape of ViT-related models. The vertical layout separates CNN-based methods (top), hybrid architectures (middle), and Transformer-based approaches (bottom). Border colours indicate the model group (Generative, Supervised, Contrastive, Multimodal, Mixed), while background colours denote the main institution associated with each model. Shapes encode the model type: a triangle for new architectures, an oval for regularisation-based methods, a square for knowledge-based models, and a hexagon for generic foundational models. Dot colours represent the year of introduction, and arrows highlight conceptual or architectural dependencies between models.

been proposed, leveraging the strengths of both global and local feature representation to improve performance on smaller datasets (see Fig. 2).

Several surveys (Han et al., 2022; Liu et al., 2023; Yang et al., 2022) and benchmarking studies have examined ViTs for vision tasks. However, the majority of these surveys focus on either general-purpose ViT models (Liu et al., 2023), a narrow set of ViT architectural variants, or task-specific adaptations such as detection or segmentation. As a result, these surveys fail to provide a unified comparison across different ViTs families, under consistent training protocols, and to identify which regularisation strategies are best suited for each case. Moreover, many reviews remain conceptual and do not critically analyse empirical behaviour across datasets or the practical challenges of training and deploying these models. This leads to fragmented evaluations and limited insight into how architectural choices interact with optimisation and regularisation in practice.

Despite the rapid development of ViTs, there is still no comprehensive, experimentally grounded benchmark that evaluates pure transformers, hybrid ViTs–CNN models, and classical CNNs under consistent training conditions using datasets from different domains (Takahashi et al., 2024). Existing surveys rarely address key concerns related to data efficiency, robustness, or training stability (Steiner et al., 2021). To address these limitations, this work provides a systematic benchmark across multiple datasets, offering a unified comparison that highlights strengths, limitations, and the trade-offs of practical model choices. In addition, we introduce a set of model add-ons and analyses framed around research questions targeting performance, data efficiency, and optimisation stability.

This survey evaluates multiple model families under controlled settings to identify architectural components and parameter choices that ensure robust, consistent behaviour while highlighting practical strengths and limitations. We review recent ViT developments, analyse key architectural variations and complementary components addressing their constraints, and assess performance across small, medium, and large datasets to evaluate generalisation and stability. Specifically, this paper provides:

- A comprehensive analysis of ViTs and hybrid CNN–ViT architectures, addressing training challenges such as reliance on large-scale pre-training and computational complexity.
- Identification and characterisation of hybrid models that integrate convolutional locality with transformer-based global representations, improving both performance and data efficiency across diverse visual tasks.

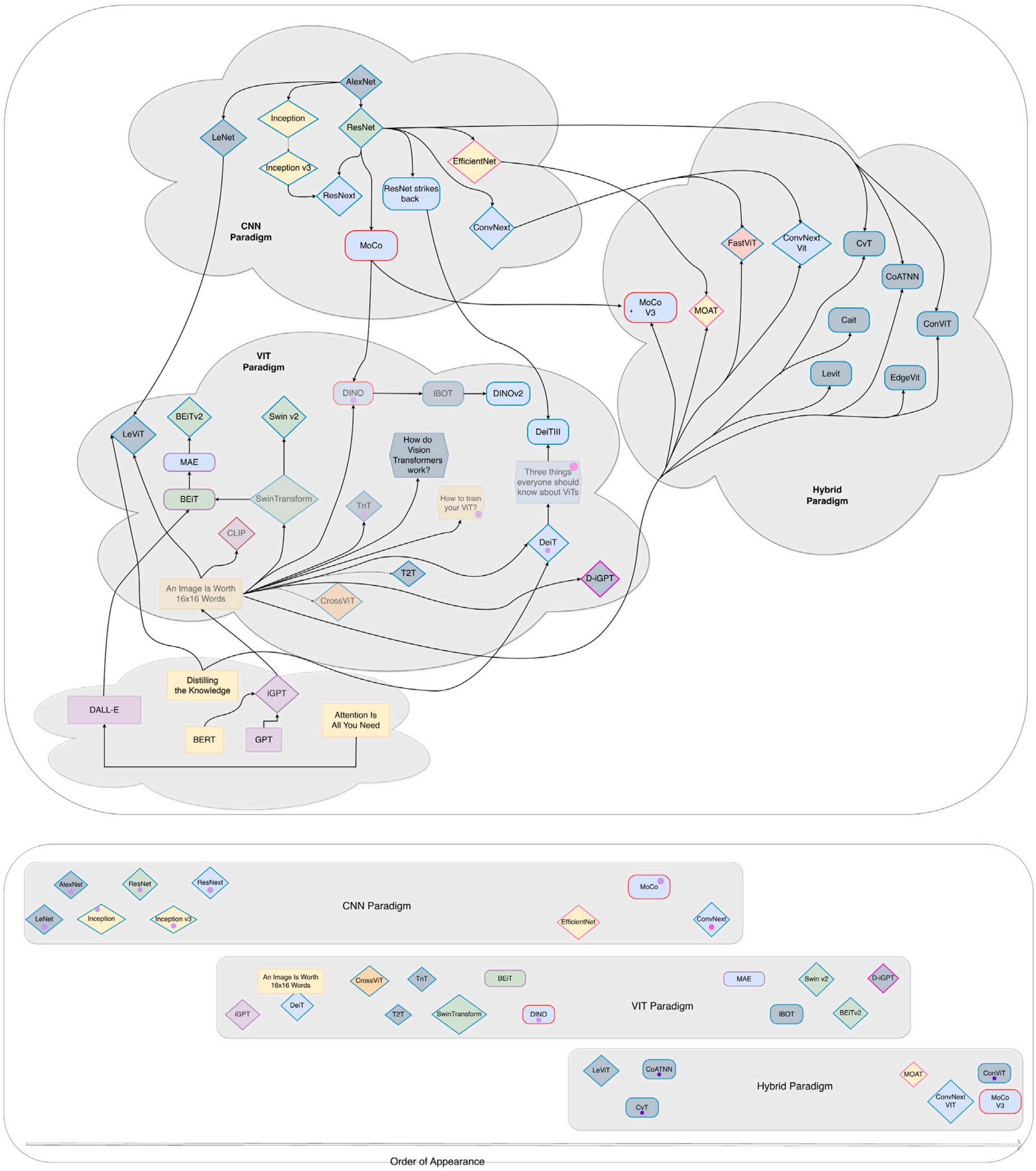

Fig. 2. Complementary views of the vision model landscape. **Top:** Clustering of representative vision models according to the three main architectural hierarchies: CNN-based models, hybrid CNN–Transformer architectures, and pure Transformer-based approaches, highlighting shared design principles and inductive biases. **Bottom:** Timeline of representative vision models ordered by year of appearance, illustrating the historical progression from early convolutional architectures to modern hybrid and Transformer-based models and highlighting key paradigm shifts in visual representation learning.

- A unified taxonomy and benchmark experiments for fair, structured comparison among ViTs variants and traditional CNNs, highlighting trade-offs between performance, complexity, and complementary components designed to mitigate limitations.
- Insights into architectural modifications and design trends that enhance usability, performance, data efficiency, and optimisation stability, guiding comparative evaluation of modern ViT-based models.

Overall, this work presents an extensive review and benchmarking of ViTs and hybrid CNN–ViT models against established CNN counterparts, using a clear taxonomy to categorise key approaches, architectural features, and comparative performance across multiple datasets.

The remainder of this paper is organised as follows: Section 2 presents the literature review and discusses related works; Section 3 describes the proposed taxonomy and previous surveys in the field; Sections 4 to 6 details the retrieved models on each category and presents the benchmarking results; Section 9 identifies the limitations of the study; finally, Section 10 concludes the study and outlines potential future research directions.

2. Literature research

The evolution of computer vision architectures began with foundational CNNs such as AlexNet (Krizhevsky et al., 2012), VGG (Simonyan & Zisserman, 2015), and ResNet (He et al., 2015), which established deep hierarchical feature extraction. Innovations like residual connections effectively addressed the vanishing gradient problem, enabling the stable training of very deep networks. InceptionNet (originally GoogLeNet) (Szegedy et al., 2015), introduced for the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2014, extended this paradigm by using inception modules, which apply multiple parallel convolutional filters of varying sizes to capture features at different spatial scales while keeping computational costs manageable. Despite these advances, CNNs remain constrained by local receptive fields, requiring substantial depth to encode global context and relying heavily on fixed inductive biases, which can limit scalability across diverse tasks.

The introduction of ViTs (Dosovitskiy et al., 2020) shifted the paradigm from convolutional hierarchies to global self-attention, enabling long-range dependency modelling, high representational capacity (Jiang et al., 2021), and scalable training. Subsequent models, such as DeiT (Touvron, Cord, & Jégou, 2022) and Swin Transformer (Liu et al., 2021), have improved data efficiency through distillation and hierarchical or windowed attention, achieving competitive results even with limited data. The modularity of transformers further supports broad downstream applications, as pre-trained backbones can be adapted for tasks such as detection and segmentation (Touvron et al., 2021).

Nonetheless, ViTs face distinct challenges: they are highly dependent on large-scale pretraining, sensitive to initialisation and optimisation schemes, and often unstable when trained from scratch. This has motivated research into efficient training, architectural regularisation, and hybrid models that integrate convolutional inductive biases (Vasu et al., 2023; Xiao et al., 2021). Lightweight and resource-efficient transformer variants have further emerged for deployment in constrained environments.

Hybrid CNN–ViT architectures combine the locality and strong priors of CNNs with the global reasoning capabilities of transformers, improving data efficiency, spatial reasoning, and optimisation stability. These hybrid approaches have demonstrated strong performance across various domains, including medical imaging (Kim et al., 2025), defect detection (Li et al., 2022), and handwriting recognition (Ben Noureddine, 2025), often outperforming both pure transformer and pure CNN models under limited data.

Although several surveys review ViT developments (Han et al., 2022; Liu et al., 2023; Yang et al., 2022), they typically emphasise individual architectural trends or task-specific adaptations rather than providing a unified comparison across pure transformers, hybrids, and classical CNNs. Moreover, existing benchmarks often suffer from inconsistent methodologies, a narrow domain focus, or simplistic complexity metrics, which limit insight into real-world efficiency and generalisation. ConVision (Bangalore Vijayakumar et al., 2024) highlights this need by proposing a unified cross-paradigm evaluation framework.

Despite rapid model evolution, a comprehensive, experimentally grounded benchmark directly comparing CNNs, pure ViTs, and hybrid CNN–ViT models under consistent training protocols remains scarce (Takahashi et al., 2024). Data efficiency, robustness, and training stability are likewise underexamined (Steiner et al., 2021). This work addresses these gaps by providing a systematic evaluation across multiple datasets, highlighting architectural strengths, limitations, and trade-offs, and motivating research directions centred on performance, data efficiency, and stable optimisation.

2.1. Previous surveys

Existing surveys have examined CNN-based methods across various imaging modalities, from traditional object recognition to medical domains such as MRI, CT, and histopathology, evaluating performance and architectural variations. However, they often overlook emerging transformer-based models or hybrid CNN–ViT approaches, and rarely discuss their potentialities or limitations.

Recently, ViTs have emerged as powerful alternatives for medical image classification, leveraging self-attention to capture long-range dependencies. While several surveys review ViT applications across modalities, they generally lack comparative benchmarking and integration with CNN-based methods.

Tables 1 summarise the most relevant surveys on CNNs and ViTs, highlighting key characteristics, benchmarking strategies, and limitations. This combined overview provides a foundation for understanding both traditional and emerging architectures in medical imaging.

Overall, from the identified surveys, CNNs still provides a strong foundation for image classification tasks, excelling at capturing local spatial patterns across modalities, such as medical imaging, as some works in Table 1 describe. In contrast, according to previous surveys, ViTs are positioned as a complementary approach to provide enhanced representation learning. While existing ViTs surveys document these advances, they often lack systematic benchmarking against CNN baselines. This genre of insights needs to be integrated in the form of benchmarking and component discussion in a holistic manner, which considers CNNs, ViTs, and hybrid CNN–ViT architectures in a side comparison to assess performance, generalisation, and computational efficiency in medical imaging.

3. Methodology

To ensure a rigorous and reproducible evaluation of ViT models, their variants, and hybrid architectures, we combine a structured literature survey with controlled experimental benchmarking.

3.1. Survey methodology

The survey traces the evolution of ViT architectures and their key innovations through a systematic review of top-tier venues (CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, TPAMI, IJCV) and major databases (Google Scholar, IEEE Xplore, arXiv, PubMed) from 2020 to the present. The list of works to include and exclude is listed in Table 2.

For each model, we identify the main architectural design (patch size, attention type, hierarchy, and hybridisation), training setup (pretraining method, data augmentation, and distillation), and performance metrics (Top-1/Top-5 accuracy, parameter count, and FLOPs). The extracted data were comparatively analysed to identify architectural trends, efficiency trade-offs, and performance patterns, benchmarked against three different datasets to effectively account for training details and capabilities.

3.2. Summary of retrieved works and categorisation

From 2020 onward, the number of relevant works retrieved across different architectural families is as follows. CNN architectures account for approximately 80 works, including major advances such as ConvNeXt (Liu, Mao et al., 2022) and EfficientNet-V2 (Tan & Le, 2019). Pure ViTs comprises around 121 works, covering various gls vit variants, hierarchical transformers, and efficient transformer designs; from these, 15 works were selected as fundamental for comparison and benchmarking. Hybrid CNN-ViT architectures include roughly 50 works that integrate convolutional priors with transformer attention mecha-

Table 1
Summary of key CNN and ViT surveys in image classification.

Category	Title/Author/Year	Contribution	Benchmarking	Pros/Cons
CNN	Transfer Learning for Medical Image Classification: A Literature Review (Kim et al., 2022)	Reviews transfer learning techniques with a focus on medical image classification.	Provides insights into transfer learning; lacks direct benchmarking.	Highlights transfer learning techniques; lacks benchmarking.
	Medical Image Classifications Using Convolutional Neural Networks: A Survey of Current Methods and Statistical Modelling of the Literature (Mohammed et al., 2024)	Compiles CNN methods for medical image classification, including architectures, frameworks, and datasets.	Provides statistical analysis of literature; no direct benchmarking.	Extensive literature compilation; lacks benchmarking.
	A Survey on Convolutional Neural Networks and Their Performance Limitations in Image Recognition Tasks (Rangel et al., 2024)	Comprehensive review of CNN architectures and performance limitations in image classification.	Discusses performance limitations; lacks extensive benchmarking.	Highlights CNN limitations; limited benchmarking.
ViT	How to Train Your ViT? Data, Augmentation, and Regularisation in Vision Transformers (Steiner et al., 2021)	Analyses how ViT training depends on data scale, augmentation, and regularisation; shows strong augmentation/regularisation can compensate for scarce data.	Extensive empirical comparisons across data sizes, augmentation strategies, and regularisation methods.	Highlights ViT sensitivity to data and training setup; limited architectural analysis.
	A Survey on Vision Transformer (Han et al., 2022)	Presents an overall view of ViT models for segmentation, classification, pose estimation and related tasks.	Benchmarking focused on parameter count and memory requirements.	Does not contemplate newer ViT variations; limited benchmarking on primary datasets.
	Transformers Meet Visual Learning Understanding: A Comprehensive Review (Yang et al., 2022)	Presents a broad survey of different ViT model variations.	Includes benchmarks.	Discusses model characteristics but offers limited analysis of pros/cons.
	Scaled ReLU Matters for Training Vision Transformers (Wang et al., 2022)	Shows that ViT training is highly sensitive to hyperparameters and that a scaled ReLU stem stabilises early training.	Benchmarks across activation functions and training settings.	Improves early-layer stability; limited focus beyond activation/initialisation.
	Scaling Vision Transformers (Zhai et al., 2022)	Studies hyperparameter scaling and strong L2 regularisation for large-scale ViT training, with major improvements in few-shot transfer.	Comprehensive benchmarks on large-scale training and transfer tasks.	Performance depends heavily on subtle hyperparameter choices; limited exploration of smaller datasets.
	A Survey of Visual Transformers (Liu et al., 2023)	Focuses on DETR-based models and multimodal approaches such as CLIP and DINO.	Provides benchmarking.	Emphasis is on DETR and hybrid models rather than pure ViT architectures.
	Transformer in Medical Imaging: A Survey (Shamshad et al., 2023)	Aggregates works using transformers for classification, segmentation, restoration, and report generation.	No benchmarking.	Comprehensive list of works; lacks evaluation or comparative analysis.
	A Survey of the Vision Transformers and Their CNN–Transformer Based Variants (Khan et al., 2023)	Provides an overview of different ViT architectures and hybrid variants.	No benchmarking.	Lists many works but does not benchmark or analyse architectural limitations.
	An Overview of Vision Transformers for Image Processing: A Survey (Kameswari et al., 2023)	Identifies main contributions of each ViT model and related surveys.	No benchmarks.	Does not cover recent ViT variants; provides limited analytical depth.
	Spatial Entropy as an Inductive Bias for Vision Transformers (Peruzzo et al., 2024)	Introduces SAR, a regulariser penalising spatially disordered attention, which improves performance under data scarcity.	Benchmarks SAR with varying dataset sizes.	Reduces reliance on large datasets; increases computational overhead.
	Vision Transformers for Image Classification: A Comparative Survey (Wang et al., 2025)	Summarises architectural variations of ViTs for image classification.	Benchmarks FLOPs and accuracy on ImageNet.	Does not compare models side-by-side; lacks detailed discussion of limitations.
	Meta-Learning Hyperparameters for Parameter-Efficient Fine-Tuning (Tian et al., 2025)	Shows that meta-learned hyperparameters significantly improve fine-tuning of large ViTs.	Benchmarks meta-learned vs. fixed hyperparameters across multiple fine-tuning tasks.	Improves robustness to hyperparameter sensitivity; introduces training complexity.

Table 2
Survey methodology and model selection criteria.

Category	Details
Keywords	"Vision Transformer," "ViT," "transformer in computer vision," "ViT limitations," "Self-Attention," "Masked Image Modelling," "Hybrid CNN-Transformer"
Databases	Google Scholar, IEEE Xplore, arXiv, PubMed
Publication Date	2020 to present
Types of Works	Peer-reviewed journal articles, conference papers, preprints
Focus Areas	Model architectures, pre-training strategies, hybrid designs, computational efficiency, applications beyond classification
Datasets	ImageNet-1K, CIFAR-10/100, MNIST, domain-specific medical datasets, custom benchmark datasets for multimodal tasks
Training Strategies	Supervised learning, self-supervised learning, distillation, masked image modelling, hierarchical or windowed attention, tokenisation improvements
Evaluation Criteria	Accuracy, Top-1/Top-5 performance, data efficiency, computational cost (FLOPs, parameters), inference latency, generalisation across datasets and domains
Inclusion Criteria	**Novelty:** Introduced a novel architectural mechanism; **Performance:** Achieved state-of-the-art or near state-of-the-art on ImageNet; **Degree Influence:** Served as canonical or influential baseline; **Paradigm Novelty:** Represented a new class in CNN, Transformer, or Hybrid architectures
Exclusion Criteria	**Redundancy:** Minor variants without meaningful architectural changes; **No Benchmark:** Models without documented ImageNet or benchmark performance; **Non-general:** Task-specific architectures not for general image classification; **Low Innovation:** Superseded models without substantial conceptual novelty

nisms, of which 10 were selected due to their unique contributions and foundational relevance.

The selected ViT models and hybrid CNN-ViT variants were chosen based on how they build on the **foundational ViT architecture** (Dosovitskiy et al., 2020), addressing key limitations of both traditional CNNs and early transformers in computer vision.

- **Why these ViT models:**
  - **ViT** replaces convolutional hierarchies with global self-attention, enabling the modelling of long-range dependencies, but it requires large datasets and is computationally expensive.
  - **CrossViT** introduces multi-scale paths, improving contextual feature extraction and better handling objects at different scales.
  - **FastViT** focuses on computational efficiency, reducing inference cost for practical deployment.
  - **Swin Transformer** implements hierarchical windowed attention, reducing the $O(n^2)$ complexity of standard self-attention and supporting high-resolution images efficiently.
  - **DeiT** enhances data efficiency using distillation, making transformers competitive with CNNs on smaller datasets.
- **Why hybrid CNN–ViT models:**
  - Hybrid models such as **CoAtNet, CvT, and ConViT** combine the **local feature extraction strengths of CNNs** with the **global attention capabilities of transformers**.
  - CNN stages or convolutional priors in these architectures improve early feature extraction, optimisation stability, and data efficiency.
  - They bridge the gap between purely convolutional or purely transformer-based designs, balancing **locality and global reasoning**, and often improving interpretability and robustness on smaller or domain-specific datasets.
- **Connections among architectures:**
  - The architectures show an **evolutionary chain**: from CNNs (AlexNet, ResNet) → ViT → ViT variants → hybrid CNN–ViT models.
  - Each step addresses specific issues: CNNs provide strong inductive biases but limited long-range modelling; ViT enables global context but struggles with data efficiency and computational cost; hybrids combine the strengths of both, maintaining accuracy and efficiency.
  - The relationships among these models reflect **incremental innovation**, where each variant extends foundational ideas (Attention is All You Need, ViT) to address practical challenges like scalability, efficiency, hierarchical feature representation, and training stability.

In short, these ViT and hybrid variants were selected because they **systematically address limitations in prior architectures**, each contributing specific improvements in scale, efficiency, locality/global feature balance, while maintaining connections to the original ViT design principles. Table 3 identifies the works and main characteristics. The models listed reflect major milestones that either introduced a new architectural paradigm or significantly advanced performance and scalability on ImageNet.

Table 3 categorises the most promising works that were selected based on defined criteria and briefly summarises the retrieved models. Fig. 1 provides a visual overview of the primary relationships and dependencies among these models. At the centre of the diagram, ViT (Dosovitskiy et al., 2020), inspired by Attention is All You Need (Vaswani et al., 2017), serves as the foundational transformer-based vision model. ViT model variants such as CrossViT (Chen et al., 2021), FastViT (Vasu et al., 2023), and Swin Transformer (Liu, Hu et al., 2022) extend the foundation work of Dosovitskiy et al. (2020): CrossViT introduces multi-scale paths for constructing richer contextual features, FastViT (Vasu et al., 2023) focuses on improving the computational efficiency for practical deployment, and Swin Transformer (Liu, Hu et al., 2022) implements hierarchical windowed attention to reduce the $O(n^2)$ complexity of standard self-attention.

These innovations are depicted in the diagram of Fig. 1 as iterative extensions of the original ViT, highlighting the adaptability and evolution of transformer architectures in the context of vision tasks. The diagram also situates these models in a broader context of computer vision evolution where foundational CNNs models such as AlexNet (Krizhevsky et al., 2012), and ResNet (He et al., 2015) have established themselves as relevant hierarchical feature extraction models, providing strong spatial inductive biases, with innovations like residual connections enabling stable optimisation of deep networks.

Table 3
Summary of retrieved models grouped by architectural families, filtered according to inclusion criteria.

Model/Year	Key characteristics	Contribution/Notes
CNN-based Models (Filtered by Inclusion Criteria)		
ResNet (He et al., 2015)	Residual blocks to address vanishing gradients	Good performance on standard benchmarks; increased training time and large model size, but remains a strong baseline.
RegNetY/RegNetZ (Radosavovic et al., 2020)	Structured design space; group convolutions; predictable parameter–FLOP tradeoffs	Efficient and scalable; strong Top-1/Top-5 accuracy on benchmark datasets; slightly behind modern CNNs.
ResNet-RS (Bello et al., 2021)	Enhanced ResNet with EMA, stochastic depth, improved regularisation and modern data augmentation	Stronger than classical ResNets; robust transfer to medical datasets; competitive with state-of-the-art CNNs.
EfficientNetV2 (Tan & Le, 2019)	Compound scaling of depth, width and resolution	High accuracy and efficiency; essential baseline for early ViT comparisons.
ConvNeXt (Liu, Mao et al., 2022)	CNN redesigned using Transformer-era architectural principles	Modern CNN achieving ViT-level ImageNet accuracy; establishes a new architectural class.
Transformer-based (ViT) Models		
ViT-Base (Dosovitskiy et al., 2020)	Patch embedding + global self-attention	First pure vision transformer; highly scalable with large datasets.
iGPT (Chen et al., 2020)	Autoregressive transformer; self-supervised	Early transformer for images; limited performance but conceptually influential.
DeiT (Touvron et al., 2021)	Distillation token; efficient training	Competitive results without large datasets; strong data-efficient baseline.
CrossViT (Chen et al., 2021)	Multi-scale transformer branches; feature fusion	Improves multi-scale representation learning.
T2T-ViT (Yuan et al., 2021)	Progressive tokenisation (Tokens-to-Token)	Improves local structure modelling at the cost of complexity.
TNT (Han et al., 2021)	Nested transformers for inner and outer tokens	Hierarchical token representation; heavy architecture.
Swin Transformer (Liu et al., 2021)	Shifted-window attention; hierarchical design	Computation-friendly and high-performing for high-resolution tasks.
BEiT (Bao et al., 2021)	Masked image modelling using tokenizer-based reconstruction	Introduced MIM for vision transformers; strong pretraining strategy.
MAE (He et al., 2022)	Masked autoencoder; pixel-level reconstruction	Simple, efficient MIM pretraining; outstanding fine-tuning results.
DINO (Caron et al., 2021)	Self-distillation with global/local views	State-of-the-art SSL method; strong representations.
iBOT (Zhou et al., 2021)	Contrastive + masked modelling hybrid	Strong SSL model; complex training pipeline.
SwinV2 (Liu, Hu et al., 2022)	Scaled cosine attention; post-norm architecture	Improved stability and performance over Swin.
BEiT v2 (Peng et al., 2022)	Advanced tokenisation; teacher–student MIM	Improved accuracy at the cost of more compute.
DINOv2 (Oquab et al., 2023)	Large-scale SSL with curated datasets	Excellent generalisation and transfer learning performance.
D-iGPT (Ren et al., 2023)	Distilled iGPT; improved masked modelling	More efficient and accurate than original iGPT.
Hybrid CNN–ViT Models		
LeViT (Graham et al., 2021)	Convolutional Pyramidal Pooling; Transformer Layers; Distillation Learning	Efficient architecture designed for speed
MOAT (Yang et al., 2023)	Inverted Bottleneck; Depth-Wise Convolutions; Residual Transformers	Mobile-friendly hybrid with strong accuracy
FastViT (Vasu et al., 2023)	Decoupled CNN and attention modules; Depth-Wise Convolutions; Linear Attention	High throughput; hybrid designed for deployment
CaiT-Hybrid (Kumar & Kukreja, 2025)	Convolutional Stem; Class-Attention Transformer Blocks; Stochastic Depth; Label Smoothing	Strong convolutional stem; good generalisation; moderate efficiency
ConvNeXt-ViT Hybrid (Liu, Mao et al., 2022)	Depth-wise Convolutions; Residual Transformer Blocks; Layer Norm; Stochastic Depth	High local–global feature extraction; strong accuracy; slightly higher parameter count
EdgeViT (Pan et al., 2022)	Edge-Focused Convolutions; Transformer Encoding; Dropout	Optimised for low-resolution features; efficient design
MobileViT-B (Mehta & Rastegari, 2021)	Convolutional Stem; Mobile Attention Blocks; Linear Attention; Depth-wise Separable Convolutions	Mobile-ready; good balance between accuracy and efficiency
CoAtNet (Dai et al., 2021)	Stacked convolution + attention stages	Strong scaling behaviour and data efficiency; good combination of locality and global context
CvT (Wu et al., 2021)	Convolutional token embedding and projections	Improved locality and training stability; bridges CNN inductive biases with ViT flexibility
ConViT (d'Ascoli et al., 2022)	Gated positional attention; soft convolutional bias	Combines CNN inductive bias with attention flexibility

ViTs advanced the computer vision task by replacing convolutional hierarchies with global self-attention, which allows modelling long-range dependencies when scaled and trained effectively with large datasets. Subsequent variants, such as Data efficiency image Transformer (DeiT) (Touvron, Cord, & Jégou, 2022) and Swin Transformer, enhanced data efficiency and hierarchical feature extraction.

Hybrid CNN–ViT architectures, including CoAtNet (Dai et al., 2021), CvT (Wu et al., 2021), and ConViT (d'Ascoli et al., 2022) appear as an intermediate step to accommodate the feature extraction capabilities of CNN counterparts, with the long-range modelling capabilities of ViTs, by integrating convolutional priors into transformer blocks or using CNNs stages for early feature extraction. In the diagram, these hybrids occupy a bridging position, reflecting their role in combining locality and global reasoning to improve data efficiency, optimisation stability, and model interpretability.

The selected ViT and hybrid CNN–ViT models were chosen since they directly extend the foundational ViT architecture while addressing its main limitations, namely high data requirements, weak inductive biases, and quadratic attention cost. Variants such as DeiT, CrossViT, Swin, and FastViT introduce complementary improvements, including data-efficient training through distillation, multi-scale tokenisation, hierarchical windowed attention for reduced complexity, and more efficient inference. Together, these architectures represent an evolution of transformer-based vision models focused on improving scalability, efficiency, and robustness across general-purpose and specialised image domains.

Overall, Fig. 1 illustrates the progression from CNNs to ViTs and hybrid designs, showing how successive models build on the foundational ideas of Attention is All You Need (Vaswani et al., 2017) and the original ViT (Dosovitskiy et al., 2020), highlighting the relationships between architectural families, the evolution of design principles, and the integration of convolutional and transformer-based components to address practical challenges in scalability, efficiency, and stability.

3.3. Experimental benchmarking

Empirical benchmarking complements the survey through standardised experiments conducted under controlled conditions. Performance and efficiency on small- to medium-scale datasets, such as CIFAR-100 (Krizhevsky, 2009), the Endoscopic Bladder Tissue Classification dataset (Shah, 2024), and ISIC 2019 (Codella et al., 2019), enable access to cross-domain generalisation image classification (Fig. 3).

All models were evaluated on the three identity additional datasets using consistent input resolutions and patch sizes per dataset (e.g., ViT-B with 224×224 for BTC and 16×16 patches for CIFAR-100 and ISIC 2019). Hyperparameters followed the original publications, with self-supervised models assessed via linear probing or full fine-tuning. Evaluation metrics included Top-1/Top-5 accuracy, parameter count, FLOPs, and training complexity, with ablation studies examining the impact of key architectural components.

Reproducibility was ensured using publicly available implementations and pretrained weights, with all models trained under identical settings. Experiments were conducted on two identical servers, each equipped with four NVIDIA Ampere GPUs, each featuring 80 GB of memory, and supported by an Intel Xeon CPU with 80 cores and 512 GB of RAM. Training duration and hyperparameters were fully reported. All models were trained for 150 epochs with early stopping (patience 15). Model source code available on [1]

To maintain comparability with the ViT experiments, all CNNs were trained or fine-tuned using a unified pipeline in PyTorch: consistent preprocessing, standard augmentations (RandAugment, MixUp/CutMix), AdamW or SGD optimisers with cosine annealing and warmup, matched input resolutions, and dataset-specific optimisation schedules to mitigate overfitting. All models are initially trained for 150 epochs and then further trained if convergence is not satisfactory.

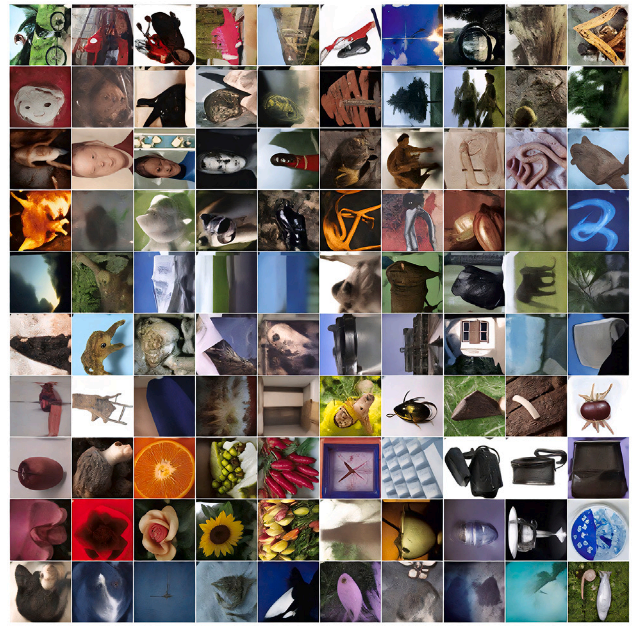

Fig. 3. Image examples.

This benchmark methodology combines systematic literature analysis with reproducible experimental evaluation, enabling quantitative, architecture-level comparisons between ViTs, hybrid CNN–ViT models, and strong modern CNN baselines for image classification tasks.[1]

4. Convolutional neural networks (CNNs)

Although recent advances in computer vision have increasingly favoured transformer-based architectures, CNNs continue to serve as competitive baselines, for example in medical imaging, where their strong inductive biases, spatial locality, and computational efficiency remain advantageous. To this end, we focus on representative CNN models developed since 2020, which provide robust reference architectures for benchmarking modern ViT variants.

While early CNNs (e.g., AlexNet, Inception, ResNet) established key principles such as hierarchical feature extraction, local receptive fields, weight sharing, and multi-scale representations, post-2020 designs extend these concepts with improved scaling rules, modernised training pipelines, and transformer-inspired components, including large kernels, inverted bottlenecks, LayerNorm, GELU activations, and AdamW optimisation schedules.

These innovations enable modern CNNs to remain highly competitive, often matching or exceeding ViT performance on small to medium datasets, reinforcing their role as essential baselines for comparative studies. Despite the growing prominence of ViTs, modern CNNs such as ResNet, EfficientNetV2 and ConvNeXt maintain strong performance while remaining computationally efficient. Using these models as reference points is essential to evaluate the relative strengths, weaknesses, and scaling behaviour of transformer-based architectures in a controlled and fair setting. Below, we summarise the selected post-2020 CNN baselines used in this work and justify their inclusion as reference models for benchmarking against ViT variants.

- **RegNetY/RegNetZ** (Radosavovic et al., 2020): systematically designed families with predictable scaling behaviour and efficient compute utilisation.

[1] https://github.com/hsouporto/Decoding_VIT

- **ResNet-RS** (Bello et al., 2021): modernised ResNet training recipe (EMA, stochastic depth, RandAugment), enabling strong performance with classic residual blocks.
- **EfficientNetV2** (Tan & Le, 2019): faster training and improved scaling (fused convs), optimised for speed-accuracy trade-offs.
- **ConvNeXt** (Liu, Mao et al., 2022): a transformer-inspired CNN that adopts inverted bottlenecks, LayerNorm, GELU, and large kernels—competitive with ViTs while preserving convolutional efficiency.
- **ResNet** (He et al., 2015) **and Reduced Bottleneck**: a modernised ResNet variant that replaces the original bottleneck with a simplified, wider residual block.

These models form the minimal set of modern baselines required to contextualise transformer-based results. Recent advances in convolutional architectures have produced a new generation of strong post-2020 CNN baselines that remain highly competitive with ViT-based models. These architectures progressively incorporate design principles inspired by transformers — such as large-kernel convolutions, inverted bottlenecks, and modern optimisation pipelines — while preserving the efficiency and inductive biases that historically characterised convolutional networks.

ResNet (He et al., 2015) appears as the next significant advancement in convolutions, competing in the same challenge and winning it in 2015. This architecture introduces the concept of residual connections (He et al., 2015), allowing an increase in the Depth of convolutional networks while tackling the vanishing gradient problem (Pascanu et al., 2012), where the effect of the training gradient vector is reduced as the number of layers in the network increases, difficulting backpropagation difficult. The residual technique involves adding the output of a deeper network layer connected to a shallower one, allowing the gradient descent to significantly affect the output of the deeper layers. This residual approach allowed significant network depth variations, designated by suffixes: **18**, **34**, **50**, and **152** layers while minimising the gradient vanishing problem. In its medium and most popular size, the ResNet-50, He et al. (2015).

ResNet with reduced bottleneck is a tailored modernised ResNet (He et al., 2015) variant that replaces the original 1×1–3×3–1×1 bottleneck with a simplified, wider 3×3–3×3 residual block. By reducing channel compression and removing the final 1×1 expansion, this design increases representational capacity, improves gradient flow, and yields stronger performance on small and medium datasets, while preserving ResNet's stability and training efficiency.

Further developments such as RegNetY and RegNetZ (Radosavovic et al., 2020) formalised a structured design space based on empirical scaling laws, enabling predictable parameter-FLOP behaviour and establishing scalable baselines with strong accuracy. Subsequent improvements were introduced by ResNet-RS (Bello et al., 2021), which revitalised classical residual networks through modern training strategies including EMA, stochastic depth, and advanced augmentation.

ResNet-RS (Bello et al., 2021) revisits the canonical ResNet architecture (He et al., 2015), disentangling the effects of architecture, training methodology, and scaling strategy. The study shows that training and scaling choices can have a larger impact on accuracy than architectural changes. Two improved scaling rules are introduced: the first prioritises depth scaling in regimes prone to overfitting (use width scaling otherwise), and the second increases input resolution more gradually than previous recommendations, being faster than EfficientNets.

A major step in efficiency-driven design was achieved with EfficientNetV2 (Tan & Le, 2019), which refined compound scaling and introduced fused convolutions to accelerate training without compromising accuracy. More recently, ConvNeXt (Liu, Mao et al., 2022) demonstrated that carefully re-engineered CNN architectures can match or exceed the performance of transformer-based models, integrating transformer-inspired components such as LayerNorm, GELU activation, and large-kernel depthwise convolutions while maintaining conventional convolutional priors. Together, these models form the dominant family of modern convolutional baselines, providing essential comparison points for evaluating ViTs and hybrid designs under controlled conditions. Table 4 summarises their methodology, performance on ImageNet-1K, and main strengths and limitations.

CNN models, such as ResNet, and EfficientNet, have shaped the evolution of deep learning in image classification. AlexNet established itself as a fundamental deep learning model associated with a revolution in computer vision by introducing the concept of deep convolutional neural networks and pioneering the inclusion of techniques such as ReLU activation and dropout regularisation to facilitate the computation of gradients and avoid overfitting, respectively.

The Residual Learning concept was introduced by the ResNet models, which incorporate skip connections to tackle the vanishing gradient problem commonly encountered in deeper networks, thereby facilitating training on small-scale benchmark datasets and achieving competitive results compared to current standards.

EfficientNet further advanced network design by employing a compound scaling method to uniformly scale network depth, width, and resolution, delivering competitive performance with fewer parameters compared to ResNet and resulting in a lower computational cost.

While CNNs have shaped the context of computer vision, with each model iteratively proposing modifications to enhance performance, scalability, and efficiency, these advancements are now being challenged by emerging transformer-based models, such as ViTs, which represent a new frontier in deep learning.

4.1. CNN performance analysis and discussion

To correctly assess CNN model capabilities, a benchmark was performed with different configurations, with the best results summarised in Table 5 across the defined general-purpose and medical imaging benchmark datasets.

Table 5 summarises the best results obtained across CIFAR-100, ISIC 2019, and the Endoscopic Bladder dataset using the selected modern CNN baselines.

The benchmarking results in Table 5 highlight a clear trend of increasing performance, largely driven by architectural improvements, regularisation strategies, and training optimisations. Key observations for the selected modern CNN baselines are summarised as follows:

- **Residual Learning and Robust Baseline** (**ResNet-RS** (Bello et al., 2021)): ResNet-RS incorporates modern training improvements — including stochastic depth, RandAugment, EMA, label smoothing, and improved initialisation — resulting in strong performance across all datasets (84.7% on CIFAR-100, 89.1% on ISIC 2019, 91.0% on Bladder). Its stable residual blocks and enhanced training recipe make it a reliable baseline for comparison with transformer-based architectures.
- **Residual Learning and Efficiency** (**ResNet-50** (He et al., 2015)): Residual connections significantly enhance generalisation, yielding strong results (84.7% on CIFAR-100, 89.1% on ISIC 2019, and 91.0% on Bladder). The reduced bottleneck variant preserves over 97% of full ResNet-50 accuracy while lowering FLOPs and memory requirements, demonstrating the viability of using small channel compression to recognise smaller objects.
- **Compound Scaling and Parameter Efficiency** (**EfficientNet** (Tan & Le, 2019)): EfficientNetV2 demonstrate the best trade-off between accuracy and efficiency, achieving up to 87.9% on CIFAR-100 and 93.0% on the Bladder dataset. The compound scaling approach ensures consistent accuracy gains with minimal growth in parameters. Stochastic depth, EMA, and advanced data augmentation stabilise convergence.

Table 4
Key post-2020 convolutional architectures are strong baselines.

Model/Year	Methodology	Reported ImageNet-1K	Pros/Cons
ResNet - (He et al., 2015)	Residual blocks to address Vanishing Gradient	76.3% Top-1, 93.2% Top-5 accuracy on Imagene1K	Increased training time; Large model size; Good performance even in today's standards.
RegNetY/RegNetZ (Radosavovic et al., 2020)	Structured design space derived from empirical scaling laws; group convolutions and predictable parameter–FLOP tradeoffs.	82.9% Top-1, 96.2% Top-5 accuracy on Imagene1K	Efficient and scalable; excellent compute predictability; slightly behind best CNN models.
ResNet-RS (Bello et al., 2021)	ResNet enhanced with EMA, stochastic depth, improved regularisation and modern data augmentation.	86.2% Top-1, 96.3% Top-5 accuracy on Imagene1K	Stronger than classical ResNets; benefits from modern training recipes; still limited by bottleneck design.
EfficientNetV2 - (Tan & Le, 2019)	Compound scaling of depth, width, and resolution	84.3% top-1, 97.1% top-5 Accuracy on Imagenet1K	High performance; Highly scalable; Requires careful hyperparameter fine tuning.
ConvNext - (Liu, Mao et al., 2022)	Inverted Bottleneck and normalisation inspired by ViT	85.1% top-1 Accuracy on Imagenet1K, 95.7 images/second	High accuracy; Efficient; Combines benefits of CNNs and transformers. Hard to finetune to smaller datasets.

Table 5
Benchmarking setup, training configurations, and Top-1 accuracy for selected modern CNN models across CIFAR-100, ISIC 2019, and the Endoscopic Bladder dataset.

Model	Dataset(s) & Top-1 accuracy	Training configuration	Regularisation/Optimisation	Remarks
ResNet-50 -(He et al., 2015)	CIFAR-100 — Top-1: 84.7% ISIC 2019 — Top-1: 89.1% Bladder — Top-1: 91.0%	SGD with momentum 0.9; batch size 256; cosine learning rate decay from 0.1 over 150 epochs; input 224 × 224.	Label smoothing (0.1).	Strong baseline; high stability with modern regularisation; excellent transferability to medical domains.
ResNet-50 (Reduced Bottleneck)	CIFAR-100 — Top-1: 83.5% ISIC 2019 — Top-1: 88.3% Bladder — Top-1: 90.2%	SGD with momentum 0.9; batch size 128; initial learning rate 0.05; cosine annealing; 150 epochs; bottleneck channels reduced by 50%.	Batch Normalisation; label smoothing (0.05); weight decay 1×10^{-4}.	Lightweight and efficient; retains most accuracy of ResNet-50 with lower FLOPs and memory usage. First convolution layer modified to avoid 32x reduction factor and boost performance in smaller images objects.
RegNetY/RegNetZ (2020) (Radosavovic et al., 2020)	CIFAR-100 — Top-1: 83.2% ISIC 2019 — Top-1: 88.5% Bladder — Top-1: 90.1%	SGD with momentum 0.9; batch size 256; initial learning rate 0.1 with cosine annealing; 150 epochs; input resolution 224 × 224.	RandAugment, label smoothing (0.1), stochastic depth, exponential moving average (EMA).	Efficient and scalable baseline; strong transfer performance on medical imaging tasks; maintains stable training with modern regularisation.
ResNet-RS (modern recipe) (Bello et al., 2021)	CIFAR-100 — Top-1: 84.7% ISIC 2019 — Top-1: 89.1% Bladder — Top-1: 91.0%	SGD w/ momentum 0.9; batch 256; cosine decay from 0.1; 150 epochs; input 224 × 224.	RandAugment, label smoothing (0.1), stochastic depth, EMA.	Robust baseline; excellent transfer to medical tasks.
EfficientNetV2 - (Tan & Le, 2019)	CIFAR-100 — B0: 86.7%/B4: 87.9% ISIC 2019 — B0: 90.4%/B4: 91.2% Bladder — B0: 92.0%/B4: 93.0%	RMSProp ($lr = 0.016$, momentum=0.9); exponential decay; batch size 128; input resolutions 224–380 px.	Stochastic depth; dropout (0.2–0.4); RandAugment; MixUp; label smoothing; EMA (Exponential Moving Average).	Excellent accuracy-efficiency trade-off; requires careful tuning for stable convergence on small datasets.
ConvNeXt-T - (Liu, Mao et al., 2022)	CIFAR-100 — Top-1: 87.5% ISIC 2019 — Top-1: 91.5% Bladder — Top-1: 93.4%	AdamW optimiser ($lr = 4 \times 10^{-3}$, weight decay 0.05); cosine schedule with warmup (20 epochs); trained for 150 epochs; input 224 × 224.	Layer scaling (1×10^{-6}); stochastic depth (0.1); MixUp (0.8); CutMix (1.0); label smoothing (0.1).	Bridges CNN with transformer training paradigms; strong regularisation provides stability; high performance with lower inference footprint.

- **Modernised Convolutional Design (ConvNeXt-T (Liu, Mao et al., 2022))**: ConvNeXt-T achieves the highest overall accuracy (87.5%, 91.5%, and 93.4% across CIFAR-100, ISIC 2019, and Bladder). It integrates transformer-inspired components—such as Layer Normalisation, inverted bottlenecks, and large kernel convolutions—bridging the ingredients of CNN and ViT. In combination with AdamW optimiser, cosine scheduling, and weight regularisation, ConvNeXt demonstrates training stability.
- **Cross-Domain Generalisation:** Across all models, medical datasets (ISIC 2019 and Bladder) exhibit higher Top-1 accuracies than CIFAR-100, primarily due to their lower intra-class variability. Architectures like ResNet and ConvNeXt show superior adaptability, while regularisation techniques such as MixUp and RandAugment can mitigate overfitting on limited data.

Overall, these results trace an evolutionary trend in modern CNNs—from residual networks with enhanced training (ResNet-RS), to compound-scaled architectures (EfficientNetV2), and finally to hybrid CNN–Transformer designs (ConvNeXt). These architectures provide strong, well-regularised baselines for benchmarking ViTs and hybrid CNN–ViT models, highlighting the continuing relevance of CNNs in both general-purpose and medical image classification tasks.

5. Vision transformers - ViTs

In the advent of the Natural Language Processing (NLP) revolution (Vaswani et al., 2017), the original NLP Transformer architecture (Vaswani et al., 2017) has been adapted for vision-related tasks.

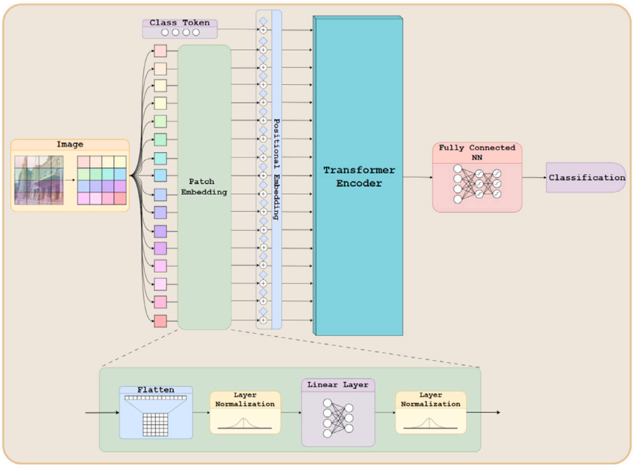

Fig. 4. ViT diagram of token projections.

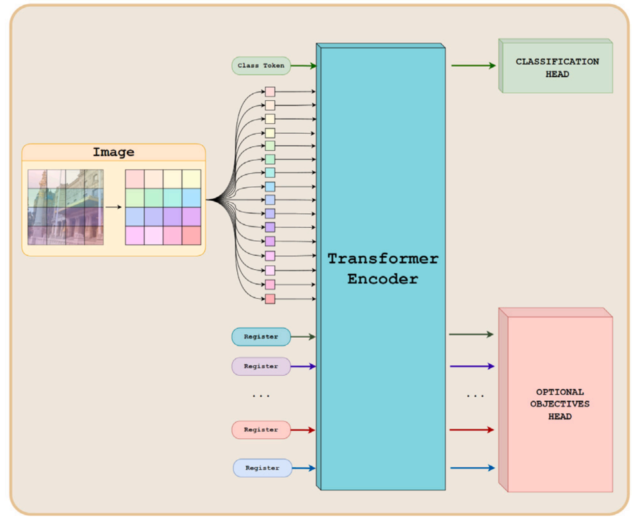

Fig. 5. Generic multiple register architecture.

The key Transformer innovation relies on the Self-Attention (SA) mechanism, which allows for capturing complex dependencies across long sequences, surpassing the limitations of a fixed-size window used in traditional approaches, such as gls cnn and Long-Short Term Memory (LSTM) models. The SA measures the degree of influence of one token on other tokens within a window, which can have an infinite size (limited by memory footprint). The architecture not only revolutionised NLP but also demonstrated remarkable efficacy in image processing, given its capacity to deal with spatial data, a fact underscored by the introduction of the initial work on ViT (Dosovitskiy et al., 2020) and Image Generative Pre-trained Transformer (iGPT) (Chen et al., 2020) with great success.

To address the scalability challenge posed by the Transformer's $O(n^2)$ quadratic parameter growth, a modification to the original softmax (Bridle, 1989) SA pooling (Vaswani et al., 2017) is proposed by Li et al. (2020) combined with the use of linear SA (Li et al., 2020) to overcome the $O(n^2)$ problem.

To avoid pixel-level interactions in both ViT and iGPT models, a ViT modification that employs a patch-based approach is proposed Dosovitskiy et al. (2020) and Chen et al. (2020). The input image is split into equal-sized patches that are projected into flattened token representations and included as learnable sequential tokens, as shown in Fig. 4. In contrast, iGPT employs a more straightforward technique, reducing the image resolution to a predefined minimum and flattening it before feeding it to the model encoder.

While ViT have garnered significant traction, spurred by their scalability and effectiveness, iGPT has inspired the appearance of new regularization techniques that have shown significant improvements to the ViT architecture: Building upon the achievements of the GPT (Radford et al., 2018) the iGPT model introduces a generative self-supervised learning technique, by masking image tokens to be predicted by a Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) or GPT visual tailored model, trained in a Masked Image Modeling (MIM) fashion similar to Masked Language Modeling (MLM) used in NLP.

In the original ViT work (Dosovitskiy et al., 2020), a different-sized transformer with varying input sizes and encoder-block quantities, identified by the following letters (**S**)mall, (**B**)ase, (**L**)arge and (**H**)uge, sorted respectively by parameter count. Transformers may also use different patch sizes during the embedding phase. For this reason, the typically selected values are 16, 8, and 4, allowing for easy calculation of an integer number of patches. For the sake of a fair comparison between networks, the accuracy reported in this work and the following subjects will refer to, when available, the ViT-B with 384 × 384 resolution and 16 × 16 patches, with this specific architecture highlighted in cases where information is missing. The ViT-B is the most famous architecture from this family, being comparable to ResNet-50 (He et al., 2015), although it has almost four times more trainable parameters (86M vs. 23M).

ViTs presents almost a match to the one reported for ResNet-50 in term of ImageNet performance (He et al., 2015), and nearly 10% for the same network when trained at (Sun et al., 2017) dataset. On the other hand, the iGPT performance is slightly lower (Chen et al., 2020) on the same dataset.

Building on the foundations established by ViTs, DeiT emerges as the next generation of architectures harnessing Transformers, featuring refined structures specifically designed for image classification via distillation learning.

Distillation Learning, proposed by Hinton et al. (2015), distils insights from a complex "teacher" model to a more straightforward "student" model. This process entails compressing the information from the larger teacher model into a simpler one. The process starts by defining the student network's learning objective, encompasses not only the typical object/class labels but also the prediction vector of the teacher network. Essentially, the student network learns to replicate the output results of the teacher network, capturing internal data relationships as a "shortcut" to the teacher's projected data. Typically, students learn official labels (represented by one-hot encoded vectors) through standard loss functions to guide self-supervised learning. However, softer insights from the teacher are acquired through less punitive loss functions. Distillation Learning can be applied to various model types, including CNNs.

For ViT, however, Distillation Learning regularisation proves particularly useful due to the presence of learnable tokens used explicitly for target prediction. Additionally, in Touvron et al. (2021), a new parameterised token is proposed for learning the soft label from a teacher model, in parallel with the class token for learning the hard label. This enhancement led to a reported (Touvron et al., 2021) accuracy of 83.1%, approximately 4 p.p. better than the reported results of the original ResNet-50 (He et al., 2015) and 6 p.p. (Dosovitskiy et al., 2020) higher than the original ViT trained on the same dataset. Additionally, the DeiT architecture is faster during training and requires less training data.

The DeiT also provided a fresh perspective on ViT and their capability to address multiple downstream tasks at once (Darcet et al., 2023; Touvron, Cord, El-Nouby et al., 2022), with the generic idea of this concept summarised on Fig. 5.

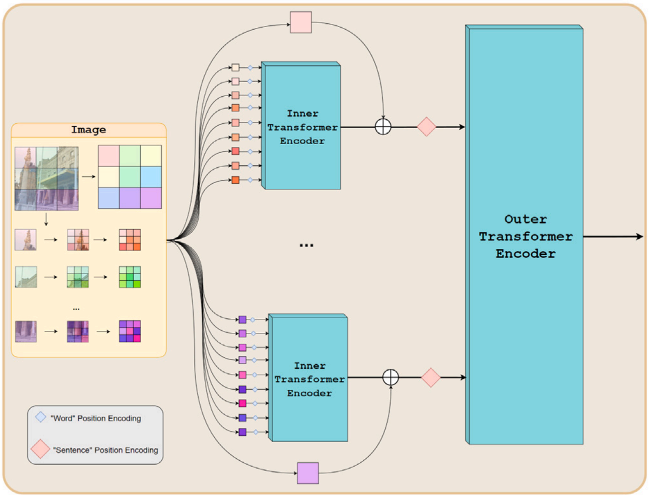

Fig. 6. A depiction of the TNT architecture (Han et al., 2021).

The DeiT architecture is further explored in Touvron, Cord, and Jégou (2022), which closely follows the new advancements on Residual Neural Network (ResNet) (Wightman et al., 2021). New augmentations and regularisation techniques were introduced by the name of DeiTIII, increasing the top-1 accuracy by 2 p.p. (top-1 accuracy of 85%)

Modifications to different models have enabled improvements to the original ViT models. One example is the CrossViT (Chen et al., 2021) architecture, which presents a new classification network designed to capture both local and global relations through novel SA layers and a Feature Fusion (Chen et al., 2021) layer. The network consists of two branches, one with larger patches and the other with smaller ones, each equipped with its own transformer encoder block. Once projected, the resulting tokens of both branches are pooled together in the so-called process of Feature Fusion through SA layers. These modifications allegedly (Chen et al., 2021) resulted in a 0.8 p.p. increase in accuracy over the original DeiT.

Tokens-to-Token ViT (T2T) (Yuan et al., 2021), Transformer in Transformer (TNT) (Han et al., 2021) and SwinTransformers (Liu et al., 2021), represent the next major achievement. These architectures share the same original patch-based approach proposed by ViT in Dosovitskiy et al. (2020). However, with the addition of a hierarchical system of patches and attention windows to preserve local structures, it avoid the lost of relevant image structures during the embedding process (Liu et al., 2021). For the T2T architecture, this system would translate into progressive tokenisation (Yuan et al., 2021), where the original tokens would be constantly re-patched and re-tokenised into smaller ones with overlapping; This process results in a more data-efficient ViT, avoiding the co-adaptation of self-attention into only given image space, achieving state-of-the-art accuracy training only on ImageNet1k (Deng et al., 2009) in opposition to JFT-300M (Sun et al., 2017) as previous architectures.

Regarding the TNT architecture, a recursive approach to hierarchical patch embedding is set in place, where the network subdivides each original patch into smaller, new patches. It projects them using "Inner Transformer Blocks" (Han et al., 2021), and the result of these smaller embeddings are then added to the generate the complete embedding representation of the original patch before being fed to the main transformer layer, capturing both local and global relations, as illustrated in Fig. 6.

On the contrary, for SwinTransformersV1 (Liu et al., 2021), the hierarchical system involves increasing the size of patches inversely, in a proportional way to the number of patches, so that 4 patches in one block correspond to 1 larger patch in the subsequent block. On the work of Touvron, Cord, El-Nouby et al. (2022), this hierarchical patch subdivision is further revisited, with the attention windows from the SwinTransformers corresponding to the grouping of patches where the corresponding attention layer is calculated. These groupings typically consist of 4 × 4 patches arranged in two ways: a standard grid, where the image is uniformly cropped (Fig. 7), or a Shifted Window, where the windows are related to non-adjacent regions in the given image. The combination of these factors provides different levels of attention with varying granularity, ranging from finer interactions with smaller patches towards the global relationships gathered in the final layer, more closely aligned with the regular operation of a ViT. The SwinTransformer architecture deviates from the original ViT taxonomy, adding (in addition to the typical hyperparameters of resolution, patch size, and transformer size) a hyperparameter for attention window size, typically set at 7 or 12 for patches of size 4.

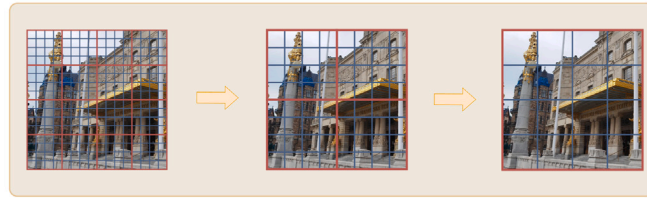

Fig. 7. Hierarchical patch and windows system from SwinTransformer (Liu et al., 2021).

SwinTransformersV2 (Liu, Hu et al., 2022) is a revision of SwinTransformerV1, aiming to address scalability issues by making adaptations to the Masked Self-Attention (MSA). Firstly, the normalisation layers were postponed after the attention calculation. Secondly, a Scaled Cosine Attention (displayed in Eq. (1)) is employed to replace the Scaled Dot-Product Attention, thereby avoiding that specific token pairs daminate final the attention maps. This modification enables the reuse of the cosine function's inherent normalisation properties (Eq. (1)) to facilitate token product scaling.

Finally, the model employs a continuous logarithmic position embedding bias, which destilates complex over easier patches, promoting window scalability. The SwintransformerV2 (Liu, Hu et al., 2022) reportedly achieves 87.1% accuracy on ImageNet1K (Deng et al., 2009), beating its predecessor (Liu et al., 2021) on roughly 2 p.p.

$$\begin{cases} Sim(q_i, k_j) = \frac{cos(q_i,k_j)}{\tau} + B_{i,j} \\ SCA = Softmax(\frac{Sim(Q,K)}{\sqrt{d_k}})V \end{cases} \tag{1}$$

BERT Pre-Training of Image Transformers (BEiT) (Bao et al., 2021) and Masked Autoencoders (MAE) (He et al., 2022) revisit the development pattern of iGPT, leveraged by the concepts of text transformers and self-supervised learning. Similar to the work of Chen et al. (2020), the BEiT proposes an MIM, which is greatly inspired by the MLM lalso used to train language models. Both architectures utilise the DALL-E (Ramesh et al., 2021) image tokeniser system to generate the initial embedding, with these embeddings then being fed into the transformer encoder for training.

Despite the network similarities, the networks differ in terms of the calculated loss and the internal architecture. BEiT (Bao et al., 2021) employs a BERT (Devlin et al., 2018) like model as an encoder and features token-level losses, whereas MAE (He et al., 2022) utilises a ViT (Dosovitskiy et al., 2020) as an image encoder and deals with pixel-level losses, relying on a decoder to translate the output tokens into the task in hands. The general idea proposed for MIM is illustrated in Fig. 8. The models achieved reported results of 84.6% (Bao et al., 2021) and 83.6% (He et al., 2022) for BEiT and MAE respectively.

Later modification to the original iGPT (Chen et al., 2020) model led to the appearance of D-iGPT (Ren et al., 2023). Unlike its original version, D-iGPT shifts towards tokenization; it sets a second training

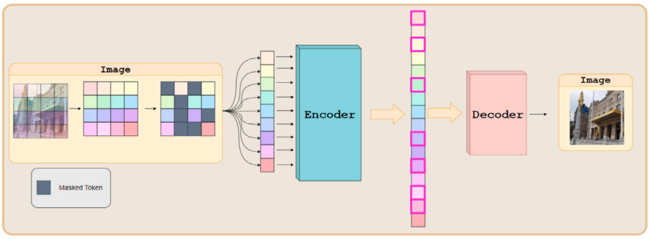

Fig. 8. Diagram representing the functionality of MIM, technique utilised by BEiT and MAE.

objective of distillation learning alongside token prediction. The model relies on a teacher model and a decoder, which, given the encoder's embeddings, minimises the distance from the teacher's predictions. This later version of iGPT achieves, according to Ren et al. (2023), 86.2% accuracy at ImageNet1K.

The following family of ViTs transformers introduces a novel regularisation system that can be utilised across multiple architectures. One example is self-**di**stillation with **no** labels (DINO) architecture (Caron et al., 2021). It consists of a self-supervised learning technique to achieve consensus between two identical models. This technique was further refined in the Image BERT pre-training with Online (iBOT) (Zhou et al., 2021) model, which introduces a more aggressive regularisation of the embedded vectors by adding extra complexity through MLM style. Reportedly, iBOT with this MLM style technique outperforms its predecessor (DINO) by about 1 p.p (79.5%) when paired with a linear classification algorithm.

After the introduction of iBOT (Zhou et al., 2021), DINO (Caron et al., 2021) model scheme was subject to further revision, exploring the same architecture as iBOT (Zhou et al., 2021), but with new techniques of regularisation and an additional curation of data. This innovation, referred to as DINOV2 (Oquab et al., 2023), brought significant increases in accuracy, reaching 84.7% using a ViT-L backbone using a self-supervised learning approach.

In similar ways as DINOV2, other model were subject to revision, namely BEiTV2 (Peng et al., 2022), SwinTransfomerV2 (Liu, Hu et al., 2022) and D-iGPT (Ren et al., 2023). The BEiTV2 extends the MIM architecture of BEiT (Bao et al., 2021), with the three core components composing the new network: the encoder-tokenizer, the decoder, and the teacher. Contrary to the previous, this version does not rely on DALL-E (Ramesh et al., 2021) as a tokenizer but rather on a proprietary codebook of token representations that, given a masked image, the encoder evaluates patches and projects vectors that are subject to exchange through the nearest neighbour, for the tokens present in the codebook vocabulary. Given the selected tokens, the decoder outputs a feature vectors that are compared with the teacher model feature vectors to maximise the similarity between them. These modifications significantly improved (Peng et al., 2022) BEiT's (Bao et al., 2021), achieving an accuracy of 85%, a remarkable value for a self-supervised learning technique.

Although not explicitly defining new architectures, new techniques were proposed to further enhance the capabilities of the ViT. Namely, using registers (Darcet et al., 2023) by exploiting the transformer capabilities for parallel computation. The problem of the addition of multiple learnable tokens (similar to the original class token) is addressed, aiming to either "clear" the embedding of the class token or add new objectives to the transformer forward pass (as used at Touvron et al., 2021).

The RealFormer (He et al., 2021), initially proposed for language models, utilises more residual connections between Attention layers alongside post-normalisation of tokens. Finally, in a recent study (Wortsman et al., 2023), the idea of modifying the original SoftMax (Bridle, 1989) layer for a Rectified Linear Unit (ReLU) was introduced, simplifying the final image classification task.

Even though this work focuses on image classification, the growth of the ViT architecture across the Computer Vision field is worth noting. Namely, ViT-based segmentation techniques, such as Pyramid Vision Transformer (PVT) (Wang et al., 2021), **Se**gmentation **Tr**ansformer (SETR) (Zheng et al., 2021) and the previously mentioned Segmenter (Strudel et al., 2021).

A summary of the primary key aspects of each ViT model variation discussed is listed in Table 6.

In summary, recent advancements in ViT architectures demonstrate a continuous trend of improving performance through architectural refinements.

Self-supervised approaches, exemplified by DINOV2 and BEiTV2, achieve remarkable accuracy gains by leveraging enhanced regularisation, novel tokenisation schemes, and teacher-student feature alignment. Complementary techniques, such as additional learnable tokens, residual connections in attention layers, and modifications to activation functions, further enhance the efficiency and capabilities of ViT models.

5.1. VIT performance analysis and discussion

Table 7 aggregates the comparative performance of the evaluated Transformer-based vision models across CIFAR-100, ISIC 2019, and Endoscopic Bladder datasets.

Hierarchical, MIM, and distillation-based architectures demonstrate stronger accuracy and efficiency; however, they are hard to train.

From the benchmark analysis, it is evident that recent models offer advantages over early self-attention mechanisms, including hierarchical tokenisation, masked image modelling, and hybrid self-supervised–distillation strategies. Some of the key insights found are:

- **Vision Transformers (ViT)** - (Dosovitskiy et al., 2020): The ViT patch-token embedding architecture translates fixed-size image patches into sequences that are processed by a self-attention layer. It demonstrated competitive ImageNet performance (77.9% Top-1) but suffers from $O(n^2)$ complexity due to the attention computation, which hinders its use in large images and renders it highly data-dependent, requiring large-scale pretraining for reasonable convergence.
- **Image GPT (iGPT)** - (Chen et al., 2020): iGPT flattened images into 1D sequences for pixel-level prediction in an auto-regressive fashion. Despite its conceptual novelty, it achieved limited accuracy and required high computational training time, motivating later patch-based adaptations, such as SwinTransformers.
- **Data-efficient Transformers (DeiT** - (Touvron et al., 2021)): The distillation approach enables smaller student models to inherit the performance of large, pre-trained teachers. It improved accuracy to 81.8% Top-1 while significantly reducing training data requirements. However, reliance on teacher models adds complexity to the training pipeline.
- **CrossViT** - (Chen et al., 2021): Multi-scale patch embeddings with intermediate feature fusion layers throughout the network enable CrossViT to capture both local and global feature representations. Achieves a 82.6% Top-1 accuracy at the expense of increased architectural complexity and computational overhead.
- **Token-to-Token ViT (T2T** - (Yuan et al., 2021)) and **Transformer-in-Transformer (TnT, 2021)** (Han et al., 2021): Both models introduce hierarchical embeddings—T2T through progressive re-tokenisation and TnT via inner transformers per patch. They achieve up to 83.3% Top-1 accuracy, offering improved data efficiency and fine-grained feature learning at the cost of higher model complexity.
- **Swin Transformers** - (Liu et al., 2021): Shifted window attention in combination with hierarchical patch merging allows for achieving good performance. Its structured window attention scheme reduces computational cost; however, it requires a delicate set of hyperparameter tuning.

Table 6
Resume of the models using Transformers.

Author/year	Methodology	Performance	Pros/Cons
ViT-Base - (Dosovitskiy et al., 2020)	Image split into patches, Linear projection	77.9% Top-1, 93.8% Top-5 Accuracy on ImageNet1K	$O(n^2)$ complexity problem
iGPT - (Chen et al., 2020)	Image reduced and flattened, self-supervised learning	72.0% Top-1 Accuracy ImageNet1K	$O(n^2)$ complexity problem; Low performance; Dismisses Labelled data
DeiT - (Touvron et al., 2021)	ViT-Base with additional token, Distillation Learning (Hinton et al., 2015)	81.8% Top-1, 95.6% Top-5 Accuracy on ImageNet1K	Generalises better than teacher model; Requires a robust pre-trained teacher model
CrossViT - (Chen et al., 2021)	Feature Fusion	82.6% Top-1, 96.0% Top-5 Accuracy on ImageNet1K	Increased model complexity; Multi-scale feature extraction
T2T - (Yuan et al., 2021)	Progressive Tokenization	83.3% Top-1, 96.2% Top-5 Accuracy on ImageNet1K	Tokenization Improved; High computational cost
TNT - (Han et al., 2021)	Recursive Hierarchical Embedding, Inner Transformer Blocks	82.8% Top-1, 96.3% Top-5 Accuracy on ImageNet1K	Multi scaled Hierarchical structure;Brute-forced method; High parameter count and complexity
SwinTransformers - (Liu et al., 2021)	Hierarchical patch embedding, Shifted Window Attention	84.5% Top-1 Accuracy, 97.2% Top-5 Accuracy on ImageNet1K	Efficient and Scalable; Complex implementation
BEiT - (Bao et al., 2021)	DALL-E (Ramesh et al., 2021) tokenizer, MIM, Token-level Loss	84.6% Top-1 Accuracy on ImageNet1K	Requires a large data-sample; Dismisses labelled data; Good Performance
MAE - (He et al., 2022)	DALL-E tokenizer, MIM, Pixel-level Loss	83.6% Top-1 Accuracy on ImageNet1K	Efficient pre-training; Dismisses labelled data; Simple implementation
DINO - (Caron et al., 2021)	Contrastive Learning, Global and Local Views	78.2% Top-1 Accuracy on ImageNet1K	Efficient pre-training; Dismisses labelled data
iBOT - (Zhou et al., 2021)	Contrastive Learning, MIM, Combining Losses	84.4% Top-1 Accuracy on ImageNet1K pre-trained on ImageNet22K	Complex design; Computationally expensive; High performance
Swinv2 - (Liu, Hu et al., 2022)	SwinTransformer, Post-norm Attention, Scaled Cosine Attention	87.1% Top-1 Accuracy on ImageNet1K pre-trained on ImageNet22K	Improved window attention mechanism; High performance; High training stability
BEiTv2 - (Peng et al., 2022)	Token Codebook, Teacher-Student MIM	85% Top-1 Accuracy on ImageNet1K	Improved performance; Advanced tokenization; High computational cost
DINOv2 - (Oquab et al., 2023)	Contrastive Learning, MIM, Combining Losses	84.7% Top-1 Accuracy on ImageNet1K	High computational requirements; Costly sample selection
D-iGPT - (Ren et al., 2023)	Distillation Learning, MIM	86.2% Top-1 Accuracy on ImageNet1K pre-trained on ImageNet22K	High performance; High computational requirements

- **Masked Autoencoders (MAE** - (He et al., 2022)) and **(BEiT)** (Bao et al., 2021): These models pioneered masked image modelling (MIM), using self-supervised pretraining to reconstruct masked patches. BEiT and MAE reach 84.6% and 83.6% Top-1 accuracy, respectively. Their dependence on large-scale tokenisers and high pretraining cost remains a limitation.
- **Regularisation and Self-Distillation (DINO, iBOT-** (Caron et al., 2021; Zhou et al., 2021)): DINO and iBOT introduced contrastive self-distillation, which improves the stability and transferability of self-supervised learning. iBOT, with the combination of contrastive and masked image model loss, achieved 84.4% Top-1 accuracy, demonstrating the potentialities of combined loss objectives.
- **Second-Generation Models (SwinV2, BEiTv2, DINOv2, D-iGPT)** - (Liu, Hu et al., 2022; Oquab et al., 2023; Peng et al., 2022; Ren et al., 2023): Through refined scaling, attention normalisation, and tokenisation strategies, SwinV2 reached 87.1% Top-1 accuracy, while BEiTv2 and DINOv2 achieved 85.0% and 84.7%, respectively. D-iGPT (86.2%). The integration of distillation into the generative training leads to convergence stability. Collectively, these advances bridge supervised and self-supervised learning paradigms.

Overall, Transformer-based vision models have evolved from the pure ViT using simple image patch decomposition to be embedded in sophisticated hybrid systems that integrate hierarchical attention, masked image modelling, and token distillation, allowing the creation of robust image backbones via self-supervised approaches, which are practical for various computer vision tasks.

6. VIT hybrid models

As evident from the discussion, adapting images to a sequential token system for input into the Transformer encoder poses a significant challenge due to the varying sizes of images. To overcome this limitation, various modifications that utilise convolutional filters, originating from the hybrid system, detailed in the work of Park and Kim (2022), combine the two types of network paradigms, ViTs and CNNs, allowing to accommodate both high and low pass frequency feature filters.

To explore this possibility, the Vision Transformer with Efficient Convolutions and Attention (LeVIT) work (Graham et al., 2021) introduces the *grafting* technique, which combines initial residual blocks from ResNet and a reduced DeiT with an inverse proportion, that is, the more residual layers, the fewer DeiT layers. All the results showed an improvement over each of the original Networks. This fact is yet backed up by Xiao et al. (2021), where the usage of a convolutional

Table 7
Benchmarking setup, Top-1 accuracy, and key observations for Transformer-based vision models across CIFAR-100, ISIC 2019, and Endoscopic Bladder datasets.

Model	Dataset(s) & Top-1 accuracy	Training configuration	Regularisation/Optimisation	Remarks
ViT-B (Dosovitskiy et al., 2020)	CIFAR-100 — 76.5% ISIC 2019 — 81.2% Bladder — 77.0%	Adam optimiser, lr = 0.001, batch size 128, 150 epochs, input 224 × 224.	Patch embedding; standard dropout; weight decay 1×10^{-4}.	First patch-based Transformer for vision; high parameter count; benefits from large datasets; .
iGPT (Chen et al., 2020)	CIFAR-100 — 79.1% ISIC 2019 — 74.5% Bladder — 76.8%	SGD, lr = 0.01, batch size 128, 200 epochs; reduced image flattening.	Autoregressive pixel prediction; high computation; low efficiency on small datasets.	Self-supervised approach; limited accuracy when off domain of training; inspired later MIM methods.
DeiT-B (Touvron et al., 2021)	CIFAR-100 — 84.2% ISIC 2019 — 85.3% Bladder — 87.1%	AdamW, lr = 0.003, batch size 128, 150 epochs, cosine schedule.	Distillation from teacher; label smoothing; MixUp; CutMix.	Data-efficient training; improved Top-1 accuracy over ViT; robust teacher model required.
CrossViT (Chen et al., 2021)	CIFAR-100 — 81.0% ISIC 2019 — 86.0% Bladder — 87.8%	AdamW, lr = 0.003, batch size 128, 150 epochs; multi-scale patches of 1, 0.7 and 0.4 original size.	Feature fusion; hierarchical attention; stochastic depth (0.1).	Captures local and global features; slightly higher complexity; finetuning complex.
T2T-ViT (Yuan et al., 2021)	CIFAR-100 — 81.5% ISIC 2019 — 86.5% Bladder — 88.0%	AdamW, lr = 0.003, batch size 128, 150 epochs; progressive tokenisation.	Re-tokenisation; strong regularisation; weight decay 0.05.	Efficient local–global attention; data-efficient; moderate computational cost; enable to capture variance of smaller datasets.
Swin-B (Liu et al., 2021)	CIFAR-100 — 83.2% ISIC 2019 — 88.7% Bladder — 90.5%	AdamW, lr = 0.003, batch size 128, 200 epochs; shifted window attention with 0.3 overlap (344) configuration, hierarchical patches.	LayerNorm; stochastic depth (0.1); patch merging.	Hierarchical attention; high scalability; strong generalisation across multi domains.
MAE-B (He et al., 2022)	CIFAR-100 — 82.0% ISIC 2019 — 87.5% Bladder — 89.0%	AdamW, lr = 0.003, batch size 128, 400 pretraining epochs + finetuning.	Masked image modelling; pixel reconstruction loss; dropout 0.1.	Efficient self-supervised learning; requires extensive pretraining; strong performance.
BEiT-B (Bao et al., 2021)	CIFAR-100 — 86.5% ISIC 2019 — 84.0% Bladder — 89.5%	AdamW, lr = 0.003, batch size 128, 400 pretraining epochs + finetuning.	Masked image modelling; token-level loss; DALL-E tokenizer.	Self-supervised pretraining; high accuracy; higher computational cost: lower performance on small datasets.
SwinV2-B (Liu, Hu et al., 2022)	CIFAR-100 — 87.0% ISIC 2019 — 90.0% Bladder — 91.5%	AdamW, lr = 0.003, batch size 128, 200 epochs; scaled cosine attention; hierarchical patch windows.	Improved attention; post-norm layers; stochastic depth (0.1).	State-of-the-art generalisation; high stability; slightly higher training cost.
BEiTv2-B (Peng et al., 2022)	CIFAR-100 — 84.0% ISIC 2019 — 89.0% Bladder — 90.0%	AdamW, lr = 0.003, batch size 128, 400 pretraining epochs + finetuning.	Token codebook; teacher-student MIM 80% hidden; layer norm.	Improved masked modelling; strong accuracy; computationally heavy and hard to train; required more that 150 epochs to converge.
DINOv2-B (Oquab et al., 2023)	CIFAR-100 — 88.5% ISIC 2019 — 89.5% Bladder — 91.0%	AdamW, lr = 0.003, batch size 128, 200 pretraining epochs + finetuning.	Contrastive learning; self-distillation; MIM regularisation.	Enhanced self-supervised learning; strong transfer; high computation; required further epochs to fully converge
D-iGPT (Ren et al., 2023)	CIFAR-100 — 83.8% ISIC 2019 — 88.5% Bladder — 90.0%	AdamW, lr = 0.003, batch size 128, 200 epochs; distillation + MIM at 80% hidden.	Teacher-student distillation; generative token prediction; high FLOPs.	High performance; effective token distillation; computationally very intensive: required further epochs to converge.

stem[2] for a transformer-based network, achieving better accuracy than a raw architecture.

Taking this into account, the LeVIT (Graham et al., 2021) architecture, named after father networks, DeiT (Touvron et al., 2021), and LeNet (Smith et al., 2016), are presented. Like LeNet, LeVIT employs a pyramidal pooling system for feature extraction, combining convolutions and transformers to construct a condensed, meaningful image input representation through the model layers. The model incorporates early convolutional layers for extract shallow features, adding transformers to model long-range dependencies. It also explores the

[2] In the context of Neural Networks, the stem refers to the initial layers of the Network that process the raw input before feeding more fine-grained layers

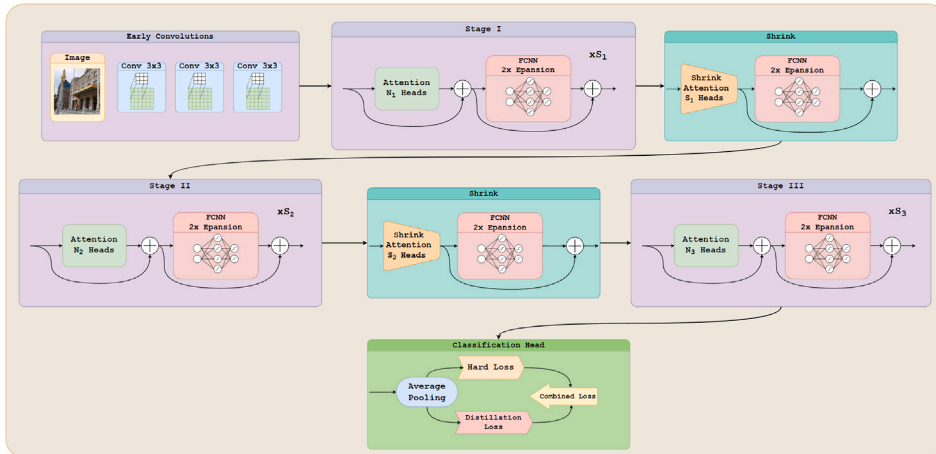

Fig. 9. Diagram representing the base template for LeViT.

same training techniques of the original DeiT, such as distilled knowledge (Hinton et al., 2015) to enhance training stability and efficiency. The basic structure of its architecture is illustrated in Fig. 9. LeViT achieves (Graham et al., 2021) 82.6% top-1 accuracy on ImageNet1K with a faster inference time, due to the main benefits of convolutions in terms of inference and training time compared to ViTs. This allows the appearance of mobile in the Network, which may vary and is denoted by efficiency.

Mobile **At**tention (MOAT) (Yang et al., 2023) follows on the success of LeVIT, aiming to minimise the network parameter while maintaining accuracy. The Network combines the inverted bottleneck presented early on in ConvNext (Liu, Mao et al., 2022), known as MobileConvs (due to its efficiency gain with no accuracy sacrifices), with the attention mechanism from transformers through residual connections. Focusing on efficiency, the convolutions are depth-wise. Given the low-resolution filters, the number of channels and blocking in the Network may vary and are denoted by the suffix number (from 0 to 5). The MOAT-2 achieves (Yang et al., 2023) a similar number of parameters to the ViT-B and the same resolution input, with 86.2% accuracy on Imagenet1K.

Aiming at mobile and low-end devices, FastViT (Vasu et al., 2023) is introduced. The network idea is to create a highly efficient hybrid architecture with State-of-the-Art (SOTA) accuracy. To achieve this effect, two separate, decoupled network modules are proposed: one for training and the other for inference. The architecture is based on a convolutional block for the initial feature extraction, followed by a mix of depth-wise convolutions (Chollet, 2016) and Linear attention-based (Li et al., 2020) transformers, also incorporating the Gaussian Error Linear Unity (GELU) (Hendrycks & Gimpel, 2016) activation function and Layer Normalisation (Ba et al., 2016). The architecture achieves (Vasu et al., 2023) 84.9% accuracy in Imagenet1K using almost half of the parameter count from the original ViT and with less inference time than the EfficientNet B6 (Vasu et al., 2023).

CaiT-Hybrid (Kumar & Kukreja, 2025) leverages a convolutional stem followed by class-attention transformer blocks. By combining stochastic depth and label smoothing, the model achieves strong generalisation on large-scale datasets, while maintaining moderate efficiency.

The ConvNeXt-ViT Hybrid (Liu, Mao et al., 2022) adopts depth-wise convolutions alongside residual transformer blocks to create strong local–global feature extraction, resulting in high Top-1 accuracy, albeit with a slightly higher parameter count. Its architecture design demonstrates that careful integration of modern CNNs with transformer layers can yield competitive performance while retaining flexibility in feature representation.

EdgeViT (Pan et al., 2022) introduces edge-focused convolutions for low-resolution features, combined with transformer encoding. This approach is optimised for efficiency in resource-constrained environments, achieving reasonable Top-1 accuracy with reduced computational overhead.

MobileViT-B (Mehta & Rastegari, 2021) integrates a convolutional stem with mobile-friendly attention blocks, using depth-wise separable convolutions, and linear attention. This mixed configuration offers an excellent balance between accuracy and efficiency, making it particularly suitable for mobile or embedded devices.

The employment of convolutional layers in the early stages to efficiently capture local features, and transformer layers to model long-range dependencies, results in competitive accuracy with optimised efficiency, facilitating deployment across a range of devices, particularly mobile-ready architectures such as MOAT and FastViT.

A summary of the primary key aspects of each hybrid model variation is presented in Table 8. Clear evidence that hybrid architectures lie in their higher efficiency and lower parameter count, achieved by leveraging attention mechanisms to enhance the effectiveness of convolutional operations. By combining the strengths of both CNNs and ViTs, hybrid configurations offer promising solutions that overcome some of the limitations of individual ViT models, facilitating real-life deployment in computer vision tasks.

Hybrid configurations hold great potential for deployment of robust computer vision models in resource-constrained environments. By combining the strengths of both architectures, it is possible to explore novel solutions that address the limitations of individual models while leveraging their complementary features.

6.1. Hybrid ViT–CNN performance analysis and discussion

Table 9 summarises the performance of representative hybrid architectures (CNN + ViT) on CIFAR-100, ISIC 2019, and Endoscopic Bladder datasets.

Hybrid networks utilise early convolutions for efficient feature extraction and transformer layers for capturing long-range dependencies, while maintaining a lower model memory footprint.

In summary, the hybrid CNN–ViT approaches can be summarised as follows:

- **Efficient Feature Extraction** (**LeViT** - Graham et al., 2021): LeViT integrates convolutional pyramidal pooling with transformer layers, achieving competitive performance across different datasets. Early convolutional layers accelerate training and inference (up to 6–7×faster than similar CNN baselines) while enabling effective feature extraction from small-scale datasets.

Table 8
Resume of the models using Hybrid approaches.

Author/year	Methodology	Performance	Pros/Cons
LeViT - (Graham et al., 2021)	Convolutional Pyramidal Pooling; Transformer Layers; Distillation Learning	82.6% Top-1 Accuracy on ImageNet1K; 6.8x Faster than EfficientNetB4 on GPU	Efficient Architecture
MOAT - (Yang et al., 2023)	Inverted Bottleneck; Depth-Wise Convolutions; Residual Transformers	86.2% Top-1 Accuracy on ImageNet1K	Mobile Ready; Efficient Architecture
FastViT - (Vasu et al., 2023)	2 decoupled modules for training and inference; Depth-Wise Convolutions; Linear Attention	84.9% Top-1 Accuracy on ImageNet1K	Efficient Architecture; Half Parameter count from ViT-B
CaiT-Hybrid - (Kumar & Kukreja, 2025)	Convolutional Stem; Class-Attention Transformer Blocks; Stochastic Depth; Label Smoothing	83.5% Top-1 Accuracy on ImageNet1K	Strong convolutional stem; good generalisation; moderate efficiency
ConvNeXt-ViT Hybrid - (Liu, Mao et al., 2022)	Depth-wise Convolutions; Residual Transformer Blocks; Layer Norm; Stochastic Depth	86.0% Top-1 Accuracy on ImageNet1K	High local–global feature extraction; strong accuracy; slightly higher parameter count
EdgeViT - (Pan et al., 2022)	Edge-Focused Convolutions; Transformer Encoding; Dropout	84.0% Top-1 Accuracy on ImageNet1K	Optimised for low-resolution features; efficient; slightly lower Top-1
MobileViT-B - (Mehta & Rastegari, 2021)	Convolutional Stem; Mobile Attention Blocks; Linear Attention; Depth-wise Separable Convolutions	85.0% Top-1 Accuracy on ImageNet1K	Mobile-ready; good balance between accuracy and efficiency
CoAtNet - (Dai et al., 2021)	Stacked Convolution + Attention Stages	84.5% Top-1 Accuracy on ImageNet1K (approx.)	Strong scaling behaviour; good combination of locality and global context
CvT - (Wu et al., 2021)	Convolutional Token Embedding and Projections	83.8% Top-1 Accuracy on ImageNet1K (approx.)	Improved locality and training stability; bridges CNN inductive biases with ViT flexibility
ConViT - (d'Ascoli et al., 2022)	Gated Positional Attention; Soft Convolutional Bias	82.5% Top-1 Accuracy on ImageNet1K (approx.)	Combines CNN inductive bias with attention flexibility

- **Mobile-Ready Architecture (MoA-T-2** - Yang et al., 2023): MoA-T-2 combines inverted bottlenecks and depth-wise convolutions with residual transformer blocks. It delivers high performance with a low parameter count, making it suitable for deployment on resource-constrained devices.
- **Decoupled Training and Inference (FastViT** - Vasu et al., 2023): FastViT separates training and inference modules, leveraging depth-wise convolutions and linear attention. This design reduces the parameter count by approximately 50% relative to ViT-B while maintaining competitive performance, enabling faster inference without significant accuracy loss.
- **Balanced Convolution–Transformer Design:** Models such as CaiT-Hybrid, ConvNeXt-ViT, and CoAtNet employ a combination of convolutional stems or residual blocks with transformer layers. This design captures both local and global feature dependencies, enhancing generalisation across diverse image domains.
- **Enhanced Locality and Stability (CvT** - Wu et al., 2021): CvT introduces convolutional token embeddings and projections to improve locality in token representations. This approach enhances training stability and enables better generalisation, particularly on smaller or domain-specific datasets.
- **Gated Positional Attention (ConViT** - d'Ascoli et al., 2022): ConViT bridges the inductive biases of CNNs with the flexibility of transformer attention using gated positional attention and soft convolutional bias. It achieves a strong balance between accuracy and efficiency across both natural and medical image datasets.
- **Cross-Domain Generalisation:** Hybrid architectures consistently outperform pure CNN or pure transformer baselines on medical imaging datasets (ISIC 2019 and Bladder). The synergy between convolutional feature extraction and attention-based global modelling enables robust performance even with lower parameter counts. Regularisation techniques such as MixUp, CutMix, and label smoothing further improve stability and generalisation on limited datasets.

Overall, hybrid CNN–ViT architectures offer a compelling trade-off between accuracy, efficiency, and scalability. By combining the complementary strengths of convolutions and transformers, these models achieve robust performance across both general-purpose and medical imaging tasks, while remaining suitable for mobile and low-resource applications.

7. Real-worlds deployments

The benchmarking results across the general-purpose CIFAR-100 and both medical datasets (ISIC 2019 and Endoscopic Bladder) and the results gathered from the original work implementations highlights clear trade-offs between the retrieved CNNs, pure ViT, and hybrid CNN–ViT architectures, each offering benefits for specific real-world deployment scenarios. Modern CNNs, such as ConvNeXt (Liu, Mao et al., 2022), EfficientNetV2 (Tan & Le, 2019), and ResNet-RS (Bello et al., 2021) consistently achieve high Top-1 accuracy across all datasets, with ConvNeXt-T reaching 93.4% on Bladder and 91.5% on ISIC 2019. The CNNs possess strong spatial inductive biases, stable training behaviour, and a reduced computational footprint compared to pure ViT counterparts, making these models especially suitable for embedded environments where reliability and energy cost outweigh the need for global top-notch accuracy. CNN training stability under small data regimes also makes them practical choices for limited annotated datasets, such as medical imaging tasks, including cancer screening or skin lesion classification.

Pure transformer models, although architecturally powerful, present a more variable performance. Vanilla ViT-B models (Dosovitskiy et al., 2020) lag significantly on all datasets due to their heavy reliance

Table 9
Benchmarking setup, Top-1 accuracy, and key observations for hybrid CNN-ViT models across CIFAR-100, ISIC 2019, and Endoscopic Bladder datasets.

Model	Dataset(s) & Top-1 accuracy	Training configuration	Regularisation/Optimisation	Remarks
LeViT-B - (Graham et al., 2021)	CIFAR-100 — 79.5% ISIC 2019 — 83.5% Bladder — 85.0%	AdamW, lr = 0.003, batch size 128, 150 epochs; pyramidal pooling; residual blocks.	Distillation from teacher; MixUp; CutMix; weight decay 1×10^{-4}.	Efficient hybrid model; super faster inference; balanced convolution and transformer layers is tricky.
MoA-T-2 - (Yang et al., 2023)	CIFAR-100 — 82.0% ISIC 2019 — 86.0% Bladder — 87.5%	AdamW, lr = 0.003, batch size 128, 200 epochs; inverted bottleneck; depth-wise convolutions.	Residual attention; layer norm; stochastic depth (0.1).	Mobile-ready; low parameter count; efficient architecture with high accuracy; require extensive training due to the depth-wise convolution layers.
FastViT-B - (Vasu et al., 2023)	CIFAR-100 — 81.0% ISIC 2019 — 85.5% Bladder — 87.0%	AdamW, lr = 0.003, batch size 128, 150 epochs; decoupled training/inference modules.	Depth-wise convolutions; linear attention; GELU activation; layer norm.	Efficient hybrid; half parameter count of ViT-B; faster inference; slightly lower Top-1 than MoA-T.
CaiT-Hybrid - (Kumar & Kukreja, 2025)	CIFAR-100 — 80.5% ISIC 2019 — 84.8% Bladder — 86.5%	AdamW, lr = 0.003, batch size 128, 150 epochs; convolutional stem; transformer blocks.	Class-attention layers; stochastic depth; label smoothing.	Leveraging convolutional layers at beginning; competitive accuracy; moderate efficiency.
ConvNeXt-ViT Hybrid - (Liu, Mao et al., 2022)	CIFAR-100 — 82.5% ISIC 2019 — 87.0% Bladder — 88.5%	AdamW, lr = 0.003, batch size 128, 150 epochs; depth-wise convolutions; transformer residual blocks.	Layer normalisation; stochastic depth; MixUp augmentation.	Strong local–global feature extraction; high accuracy; moderate efficiency.
EdgeViT - (Pan et al., 2022)	CIFAR-100 — 79.8% ISIC 2019 — 83.8% Bladder — 85.5%	AdamW, lr = 0.003, batch size 128, 150 epochs; edge-focused convolution layers; small transformer encoding.	Dropout 0.1; weight decay 0.05; residual connections.	Optimised for low-resolution images.
MobileViT-B - (Mehta & Rastegari, 2021)	CIFAR-100 — 81.2% ISIC 2019 — 85.8% Bladder — 87.2%	AdamW, lr = 0.003, batch size 128, 150 epochs; convolutional stem; mobile attention blocks.	Linear attention; depth-wise separable convolutions; stochastic depth.	Optima for mobile devices; good trade-off between accuracy and efficiency.
CoAtNet - (Dai et al., 2021)	CIFAR-100 — 82.2% ISIC 2019 — 86.5% Bladder — 87.8%	AdamW, lr = 0.003, batch size 128, 150 epochs; stacked convolution + attention stages.	Residual attention; layer norm; stochastic depth.	Strong scaling behaviour; good combination of locality and global context; high accuracy.
CvT - (Wu et al., 2021)	CIFAR-100 — 81.8% ISIC 2019 — 86.0% Bladder — 87.5%	AdamW, lr = 0.003, batch size 128, 150 epochs; convolutional token embedding and projections.	Layer norm; residual attention; stochastic depth.	Improved locality and training stability; bridges CNN inductive biases with ViT flexibility.
ConViT - (d'Ascoli et al., 2022)	CIFAR-100 — 81.0% ISIC 2019 — 85.5% Bladder — 87.0%	AdamW, lr = 0.003, batch size 128, 150 epochs; gated positional attention; soft convolutional bias.	Residual connections; layer norm; stochastic depth.	Combines CNN inductive bias with attention flexibility; good balance between accuracy and efficiency.

on large-scale generic pretraining. However, advanced model variants such as SwinV2-B (Liu, Hu et al., 2022) and DINOv2 (Oquab et al., 2023) overcome this problem, achieving competitive accuracy (e.g., SwinV2-B: 90.0% on ISIC, 91.5% on Bladder) thanks to the introduction of hierarchical attention mechanisms combined with robust self-supervised pretraining. These ViT model modifications make them suitable for applications requiring strong generalisation across different imaging devices, overcoming the domain shift challenges present in many medical image devices. Nonetheless, their computational cost, due to larger parameter counts and longer training times, remains a major obstacle for deployment in mobile or embedded medical devices.

Hybrid CNN–ViT architectures provide an attractive solution, striking a balance between performance and usability. Models such as MoA-T-2 (Yang et al., 2023), CoAtNet (Dai et al., 2021), and CvT (Wu et al., 2021) combine convolutional inductive biases with the global attention provided by the transformer architecture, achieving good performance while retaining computational efficiency. For example, MoA-T-2 and CoAtNet perform comparably to Swin (Liu, Hu et al., 2022) and DeiT variants (Touvron, Cord, & Jégou, 2022), with significantly lower complexity, achieving up to 87%–88% accuracy on ISIC and 87%–88% on the Bladder dataset. This hybrid combination makes these models particularly suitable for real-time deployment on edge devices with restricted computational resources.

Throughout is not the objective of this survey. In Table 10, we summarise the model complexity of the retrieved models in terms of parameter count on the variants benchmarked and their estimated GFLOPs, to easily access model complexity and computational deployment requirements.

Overall, the benchmark results indicate that CNNs remain the most operationally robust for small imaging tasks, transformers excel in high-variability environments with strong pretraining, and hybrid models provide the best accuracy–efficiency trade-off for real-world deployment. These findings reinforce that model selection should be driven not only by accuracy, but also by deployment requirements, dataset scale, hardware availability, and the clinical importance of reliability versus flexibility.

8. Overall model paradigm comparison

Table 11 provides a structured comparison of modern CNN-based, Transformer-based, and hybrid vision architectures. It summarises representative models, their main strengths, and observed limitations, allowing a quick overview of design trends and performance trade-offs in recent literature.

Among the retrieved models, several trade-offs need to be considered when deploying Deep learning models. According to the problem and environment to be deployed, the correct choice of models can provide substantial gains in terms of usability without substantially harming performance.

9. Limitations of the study

ViT have revolutionised the field of computer vision, introducing global attention mechanisms that surpass the capabilities of traditional convolutional approaches.

Table 10
Model, citation, approximate parameter count and GFLOPs (per forward pass). Values with "~" are approximate of standard framework implementation defaults. All values assume input 224 × 224 when otherwise mentioned.

Model (citation)	Parameters	GFLOPs	Notes
ResNet-50 (He et al., 2015)	~ 25.6M	~ 4.0	Canonical 224 × 224.
RegNetY-4G (Radosavovic et al., 2020)	~ 20M	~ 4.0	Standard mid-size RegNetY variant.
ResNet-RS-50 (Bello et al., 2021)	~ 35M	~ 5.4	Values from RS-50 variant implementation.
EfficientNetV2-B0 (Tan & Le, 2019)	~ 7.1M	~ 0.4	Canonical B0 variant.
ConvNeXt-Tiny (Liu, Mao et al., 2022)	~ 28M	~ 4.5	Standard Tiny version.
ViT-B/16 (Dosovitskiy et al., 2020)	~ 86M	~ 17.6	16 × 16 patches.
iGPT-S (Chen et al., 2020)	~ 76M	~ 10.0	Small model variant.
DeiT-B (Touvron et al., 2021)	~ 86M	~ 17.6	Similar to ViT-B architecture.
CrossViT-15 (Chen et al., 2021)	~ 44M	~ 9.0	Mid-size variant.
T2T-ViT-14 (Yuan et al., 2021)	~ 22M	~ 4.8	Canonical variant in the paper.
TNT-S (Han et al., 2021)	~ 23M	~ 5.2	Standard TNT-S.
Swin-B (Liu et al., 2021)	~ 88M	~ 15.4	Swin baseline used.
BEiT-B (Bao et al., 2021)	~ 86M	~ 17.6	Same as ViT-B backbone.
MAE-ViT-B (He et al., 2022)	~ 86M	~ 17.6	Backbone-dependent (Used ViT-B).
DINO-ViT-B (Caron et al., 2021)	~ 86M	~ 17.6	SSL applied to ViT-B.
iBOT-ViT-B (Zhou et al., 2021)	~ 86M	~ 17.6	Standard backbone ViT-B.
SwinV2-B (Liu, Hu et al., 2022)	~ 88M	~ 15.5	Baseline version.
BEiT-v2-B (Peng et al., 2022)	~ 86M	~ 17.6	Same ViT-B backbone size.
DINOv2-B (Oquab et al., 2023)	~ 86M	~ 17.6	Default implementation of baseline.
D-iGPT-S (Ren et al., 2023)	~ 33M	~ 3.8	Distilled small variant.
LeViT-128S (Graham et al., 2021)	~ 5.3M	~ 0.7	Canonical mobile setting.
MOAT-0 (Yang et al., 2023)	~ 22M	~ 5.0	Mid-size baseline used.
FastViT-SA24 (Vasu et al., 2023)	~ 20M	~ 4.4	The most representative variant.
CaiT-S36 (Kumar & Kukreja, 2025)	~ 68M	~ 13.9	The most representative variant.
ConvNeXt-ViT-Hybrid-T (Liu, Mao et al., 2022)	~ 30M	~ 5.0	Standard hybrid baseline.
EdgeViT-S (Pan et al., 2022)	~ 6.7M	~ 0.9	Small variant used.
MobileViT-B (Mehta & Rastegari, 2021)	~ 5.8M	~ 1.1	Standard ViT-B used.
CoAtNet-0 (Dai et al., 2021)	~ 25M	~ 4.0	Baseline model used.
CvT-13 (Wu et al., 2021)	~ 20M	~ 4.5	Canonical model.
ConViT-S (d'Ascoli et al., 2022)	~ 27M	~ 5.4	ConViT small variant used.

Table 11
Comparison of CNN, Transformer, and Hybrid architectures, summarising representative models, strengths, and limitations across vision model families.

Family	Representative models	Strengths	Limitations
CNN-based Architectures	ResNet (He et al., 2015) RegNetY/Z (Radosavovic et al., 2020) ResNet-RS (Bello et al., 2021) EfficientNetV2 (Tan & Le, 2019) ConvNeXt (Liu, Mao et al., 2022)	* Strong inductive biases (locality, translation equivariance) * Stable training and efficient inference * Excellent accuracy–FLOPs trade-off * ConvNeXt and ResNet-RS remain competitive with ViTs	* Limited global receptive field * Scaling is less flexible than ViTs * Often require hand-crafted scaling rules * Less adaptable than attention-based architectures
Transformer-based Architectures	ViT (Dosovitskiy et al., 2020) iGPT (Chen et al., 2020) DeiT (Touvron et al., 2021) CrossViT (Chen et al., 2021) T2T-ViT (Yuan et al., 2021) TNT (Han et al., 2021) Swin/SwinV2 (Liu, Hu et al., 2022; Liu et al., 2021) BEiT (Bao et al., 2021) MAE (He et al., 2022) DINO (Caron et al., 2021) iBOT (Zhou et al., 2021) DINOv2 (Oquab et al., 2023)	* Global self-attention captures long-range dependencies * Highly scalable from small to giant models * Strong self-supervised performance (MAE, DINO, iBOT) * Hierarchical variants achieve good efficiency (Swin) * Flexible and domain-agnostic	* Quadratic attention cost for many variants * Require large datasets for stable training * Higher compute/memory footprint * Complex pipelines (tokenizers, MIM, distillation) * Less suitable for edge deployment
Hybrid Architectures	LeViT (Graham et al., 2021) MOAT (Yang et al., 2023) FastViT (Vasu et al., 2023) CaiT-Hybrid (Kumar & Kukreja, 2025) ConvNeXt-ViT Hybrid (Liu, Mao et al., 2022) EdgeViT (Pan et al., 2022) MobileViT (Mehta & Rastegari, 2021) CoAtNet (Dai et al., 2021) CvT (Wu et al., 2021) ConViT (d'Ascoli et al., 2022)	* Combine CNN locality with Transformer global context * More efficient than pure ViTs; more expressive than CNNs * Strong performance on mid-budget compute * Excellent for mobile/edge (LeViT, MobileViT, EdgeViT) * Training stability and improved sample efficiency	* More complex architectural integration * Require careful design/tuning of CNN–attention fusion * Less scalable than full-size ViTs * Sometimes slightly behind ViTs on huge datasets

While these architectures have established themselves as state-of-the-art in numerous datasets, their success is accompanied by significant limitations, such as a wide range of hyperparameter tuning, computational cost, and the need to be pre-trained on large datasets to be effective, making them computationally intensive and challenging to deploy in resource-constrained environments.

These comparisons underscore the ongoing need for architectural advancements that integrate the strengths of these paradigms while addressing their limitations. Although this study addresses the main ViT architectures and hybrid compared to CNN counterparts, many other models still exist and have been benchmarked in many other datasets that can also provide valuable insights into model performance for specific tasks or domains.

10. Conclusions

This survey aims to provide a comprehensive analysis of the evolution of ViTs architectures in the literature, a novel paradigm that has brought innovation to the field of computer vision, adding new perspectives to the development of Deep Learning.

The architectures resulting from this innovation show a significant improvement in the SOTA across many well-established datasets but at the cost of a substantial increase in the number of parameters. Moreover, the global attention mechanism introduced by these innovative techniques is a significant contribution compared to the traditionally explored convolutional techniques in this field. ViTs demonstrates immense innovation potential, especially given the relative "saturation" of convolutional networks.

However, these architectures face challenges related to complexity and efficiency that still need to be addressed for pure ViTs (without convolutions) to become more practical tools. The emergence of hybrid networks aims to strike a balance between convolutional networks and global attention mechanisms, demonstrating excellent capabilities and high efficiency, making them suitable for mobile device applications. This suggests a new research focus that warrants further development, namely, in inference time and detailed computational requirements.

ViTs represents a crucial milestone in advancing computer vision, opening new research areas beyond the convolutional niche and potentially enabling breakthroughs previously unexplored in visual tasks. Hybrid approaches emerge as a natural consequence of such an explanation, and the tradeoff in these approaches lies in striking a balance between performance and computational efficiency. This combination enhances the performance of hybrid models while reducing computational load, making them more versatile and efficient, albeit with lower long-term accuracy.

From previous surveys, it is possible to identify some limitations, namely the lack of a standardised assessment of ViT performance across datasets, which makes it difficult to compare methods fairly, and the limitations and potentialities of each model and their central relation regarding modifications. Our work addresses this gap by comprehensively comparing works in the ViT topic. Unlike previous reviews, which often lack a clear discussion of how they relate to existing literature, our survey explicitly positions itself within the current research landscape, highlighting key differences, advancements, and unresolved challenges.

We map our contributions against prior works and identify their limitations, such as the absence of discussions on multimodal approaches, weakly supervised learning, or inconsistencies in benchmarking. Doing so ensures that our review is not an isolated analysis but a critical and integrative discussion that enhances the field's understanding while setting a foundation for future research directions.

CRediT authorship contribution statement

João Montrezol: Conceptualization, Methodology, Investigation, Writing – original draft. **Hugo S. Oliveira:** Conceptualization, Methodology, Investigation, Supervision, Writing – review & editing. **Hélder P. Oliveira:** Supervision.

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Data availability

Data will be made available on request.

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